Internet Shutdowns and Software Development:

Evidence from Bangladesh

Hector Alberto Diaz Gomez

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Abstract

This paper studies how Bangladesh's nationwide Internet shutdown of July 2024 affected observable software-development activity. Using public GitHub events for 59,158 pre-active accounts associated with Bangladesh and fourteen comparison countries, I measure the daily share of eligible users with productive GitHub activity. The paper makes three contributions. First, it documents an immediate collapse. During the July 18–23 shutdown, this share averaged 2.02 percent in Bangladesh, compared with 15.28 percent for a synthetic control, an estimated effect of −13.26 percentage points (−86.8 percent). An international difference-in-differences comparison yields −13.09 percentage points, and Bangladesh is the most extreme unit in an in-space placebo test, with an exact rank p-value of 0.0667, the minimum attainable with fourteen placebo countries. Second, Git metadata reveal a small delayed-publication backlog alongside a collapse in observable author-time creation. After the approximately 15:00 UTC July 18 onset, creation of code subsequently observed publicly by August 10 falls roughly 96–97 percent relative to pre-shutdown comparison windows; backlog is about 2.5 percent of normal volume. The original calendar-cohort backlog share of 28.3 percent therefore does not imply extensive continuation of work. Third, the cumulative deficit against the original synthetic series is 8,145 active user-days during the shutdown and 3,633 by September 4 after partial offsets; restoring connectivity did not imply a uniform recovery. In a fixed sample of 20,000 Bangladeshi accounts, activity in repositories associated with finance and information recovered fastest, whereas activity associated with accommodation and food remained depressed throughout the observed window. These sectoral patterns are descriptive comparisons with each sector's own baseline, not sector-specific causal effects. All outcomes measure observable public GitHub activity rather than total software productivity. Because the shutdown coincided with severe domestic unrest, the estimates are interpreted as effects of the shutdown episode under the assumption that no other Bangladesh-specific shock produced the same abrupt break.

Keywords: Internet shutdowns; software production; GitHub; synthetic control; difference-in-differences; digital economy.

1 Introduction

Internet connectivity is an input into an increasingly large set of production processes. For software developers, the dependence is especially direct: source code can be written locally, but modern development also relies on remote repositories, package registries, code review, issue tracking, continuous integration, documentation, and communication with collaborators. A nationwide loss of connectivity therefore interrupts both the transmission of code and many complementary tasks required to coordinate and ship software.

This paper studies a sharp episode of lost connectivity: Bangladesh's nationwide Internet shutdown in July 2024. Cloudflare reported that mobile networks began disconnecting early on July 18 and that national traffic fell to near zero later that day. Fixed broadband began to return on July 23, initially for selected sectors, while mobile connectivity recovered later, on July 28 (Belson, 2024a). Internet Society Pulse records the national disruption as beginning on July 18 and lasting roughly five and a half days (Internet Society, 2024a). The event offers an unusual setting in which a digitally intensive form of work can be observed at high frequency before, during, and after a near-complete national loss of Internet connectivity.

This paper makes three contributions. The first concerns the immediate disruption: a nationwide loss of connectivity produced an unusually large collapse in observable software-development activity relative to a synthetic counterfactual built from other countries. The second concerns adaptation: Git metadata distinguish online publication from local code creation and identify a small delayed-publication backlog alongside a roughly 96–97 percent fall in observable author-time commit creation. The third concerns persistence: the original synthetic series implies an 8,145 active-user-day shutdown deficit, followed by only a partial cumulative offset through September 4; restoring connectivity did not imply a uniform recovery, and observable activity returned at different speeds across the economic sectors associated with developers' repositories. The third contribution is descriptive heterogeneity within Bangladesh rather than a set of sector-specific causal effects.

The empirical challenge is that GitHub activity is volatile and highly heterogeneous across countries and developers. I therefore construct a multi-country donor pool and use synthetic control as the primary design. The treated unit is Bangladesh, and the donor pool contains fourteen countries with different sample sizes, including 1,373 eligible Ghana-associated accounts and 446 Jordan-associated accounts. Small-country rates may be noisy; minimum-size donor rules are reserved for advisor review. The synthetic-control weights are chosen using the pre-shutdown path of the daily active-user rate. The resulting counterfactual assigns 47.6 percent weight to Pakistan, 32.5 percent to Ghana, and 19.9 percent to Kenya. During July 18–23, the share of eligible Bangladeshi users with productive GitHub activity averaged only 2.02 percent, compared with 15.28 percent in the synthetic control. The estimated effect is -13.26 percentage points, or -86.8 percent relative to the

synthetic counterfactual.

Several complementary designs support the same conclusion. An international difference-in-differences comparison yields an effect of −13.09 percentage points, and excluding national holidays changes the estimate only modestly to −13.38 percentage points. Outcome-specific synthetic controls show declines of 86.7 percent in productive activity, 90.5 percent in shipping activity, and 91.9 percent in collaborative activity. A regression discontinuity in time at the July 17 cutoff produces a smaller but still large discontinuity of roughly −9 percentage points in the daily activity rate.

The second contribution distinguishes observable online activity from underlying code creation. A Git commit can be created locally without Internet access, while the corresponding push to GitHub requires connectivity. For a large sample of public commits, I recover Git author and committer timestamps and match them to the first observed public PushEvent. In the strict sample where the commit author is the same GitHub account that performs the push, 829 of 2,932 commits authored during the July 18–23 shutdown, or 28.3 percent, are first observed in a public push on or after July 24. Using the committer timestamp yields 27.2 percent. The share of shutdown-authored commits delayed by more than 24 hours is 29.7 percent, compared with an average of 1.1 percent across six comparable pre-shutdown placebo windows of six days each. The level comparison is more restrictive: starting at 15:00 UTC July 18 leaves 1,097 commits, 96.35 percent below duration-matched pre-shutdown windows; the 752-commit backlog is only 2.50 percent of that normal volume. The six-day comparison gives a 96.75 percent decline and a 2.45 percent backlog-to-baseline ratio. These counts concern code pushed publicly by August 10, with unresolved metadata and later pushes unobserved.

The third contribution turns from the shutdown window to the restoration period. Using a fixed sample of 20,000 Bangladesh-associated accounts selected on pre-shutdown activity, and a text classifier that assigns repositories to NAICS sectors from historical README files, I trace observable push activity by the economic sector associated with the repositories in which it occurs. Restoration of connectivity did not imply an immediate or uniform recovery. Activity in Finance- and Information-related repositories began a two-week run at or above 80 percent of its pre-shutdown level by the third week after the shutdown; Education and Health recovered more gradually; and Accommodation/Food remained well below its baseline throughout the observed window. Activity in pre-existing repositories recovered far less than sector-wide activity, consistent with part of the recovery occurring through new projects. Because this analysis compares each sector with its own pre-shutdown level rather than with an untreated control, it describes heterogeneity in recovery dynamics and does not estimate sector-specific causal effects.

The findings relate to three literatures. First, they add micro-founded evidence to work on the economic costs of Internet shutdowns and, more broadly, of infrastructure outages. Existing

studies document broad effects on output, employment, investment, and economic activity (Tagat, Phokeer, and Kreitem, 2024; Gunsilius and Van Dijcke, 2023), and Allcott et al. (2016) show how firms cope when electricity becomes unavailable. Second, the paper contributes to research on Internet access as a productive input. Hjort and Poulsen (2019), for example, find that the arrival of fast Internet increased employment and firm performance in Africa, and Akerman et al. (2015) show that broadband complements skilled work. Third, the paper joins a growing literature that uses public GitHub activity as a high-frequency measure of software work. Kreitmeir and Raschky (2024) exploit Italy's temporary ChatGPT ban and construct daily GitHub outcomes; Hu et al. (2023) use open-source contributions to study lockdowns; and Wachs et al. (2022) demonstrate the usefulness of GitHub data for measuring the geography of software production.

The interpretation is deliberately narrower than 'total software productivity.' Public GitHub events omit private repositories, other hosting services, meetings, local coding that is never pushed publicly, and non-coding work. User location is self-reported rather than physically verified. Most importantly, the shutdown was implemented amid severe domestic unrest, so the design cannot mechanically separate every contemporaneous Bangladesh-specific shock from the connectivity shock. The paper therefore estimates a large collapse in observable GitHub software-production activity during the shutdown episode, and then uses the commit-timestamp mechanism to examine whether delayed connectivity is consistent with accounting for part of the observed collapse.

2 Institutional Background

2.1 The July 2024 shutdown

The shutdown followed student protests against quotas in government jobs (Belson, 2024a). The political context matters for identification because the decision to restrict connectivity was not randomly assigned. The empirical design therefore relies on the timing and severity of the network disruption, together with cross-country and within-Bangladesh comparisons, rather than assuming that the broader domestic environment was unchanged.

Available network measurements nevertheless provide a precise chronology. Cloudflare reports outages at major mobile operators beginning during the early hours of July 18, followed by a near-complete national loss of Internet traffic around 21:00 local time. Fixed broadband began to recover on July 23, initially as a limited restoration that prioritized sectors including banking, technology, exporters, outsourcing providers, and media. Some fixed-broadband providers returned later, while major mobile providers did not show substantial recovery until July 28 (Belson, 2024a). Internet Society Pulse also records a national shutdown beginning July 18 and notes the gradual return of connectivity on July 23 (Internet Society, 2024a).

The main treatment window in this paper is July 18–23 inclusive. This choice captures the six daily observations most clearly associated with the near-total national disruption while treating July 24 onward as the start of the restoration period in daily data. The distinction is necessarily approximate because restoration began during July 23 and differed across networks. For that reason, the main estimates average over the full July 18–23 window, the weekly event study reports the restoration weeks separately, and the regression discontinuity in time is used only as a robustness check.

Table 1: Institutional timeline

Date	Connectivity status	Empirical treatment
July 16–17	Selective social-media/service blocking reported before the nationwide outage (Internet Society, 2024a)	Pre-treatment warning period; July 17 Ashura holiday handled in robustness (The Daily Star, 2024)
July 18	Mobile networks disconnect; traffic falls toward zero nationally	Treatment begins
July 18–23	Near-total nationwide Internet disruption	Main total-shutdown window
July 23–27	Partial fixed-broadband restoration	Restoration / post-treatment transition
July 28 onward	Mobile connectivity begins to recover	Post-treatment

Sources: Cloudflare Radar reporting in Belson (2024a) and Internet Society Pulse (Internet Society, 2024a); the Ashura holiday is documented by The Daily Star (2024). Dates are operationalized at daily frequency for the empirical analysis.

2.2 Why GitHub activity is informative about software production

Distributed version control separates code creation from code publication. Developers can create commits locally, but a push to a remote GitHub repository requires network connectivity. Other productive actions—opening or merging pull requests, reviewing code, closing issues, publishing releases, and commenting on collaborative work—are even more directly online. This distinction makes software development useful for studying an Internet shutdown: public GitHub activity measures an observable online margin, while commit metadata can reveal whether some code was created locally and uploaded later.

GH Archive records the public GitHub event stream and makes it available through hourly archives and Google BigQuery (GH Archive, n.d.). The data include events such as PushEvent, PullRequestEvent, PullRequestReviewEvent, IssuesEvent, IssueCommentEvent, and ReleaseEvent. For historical PushEvents, the event payload contains the number of commits and a list of commit objects, subject to a 20-commit timeline-event cap documented by GitHub. I account for this limitation in the mechanism analysis by defining a clean sample of fully observed pushes and by reporting robustness to progressively smaller push sizes (GitHub, n.d.).

3 Related Literature

Internet access as a productive input. A first literature treats connectivity as an input into production rather than a consumption amenity. Hjort and Poulsen (2019) exploit the arrival of submarine Internet cables in Africa and find increases in employment, especially in higher-skill occupations, together with evidence of firm entry, productivity, and exporting. Akerman et al. (2015) use the rollout of broadband in Norway and find that broadband adoption in firms improved the labor-market outcomes and productivity of skilled workers while substituting for unskilled workers in routine tasks. Software development is a particularly connectivity-dependent form of such skilled, nonroutine work.

Infrastructure outages and Internet shutdowns. A second literature studies what happens when an infrastructure input becomes unavailable. Allcott et al. (2016) estimate that electricity shortages reduce the revenues and producer surplus of Indian manufacturing plants, but that productivity losses are smaller because many inputs can be stored during outages, and that plants without generators are affected more severely. Their emphasis on coping margins parallels this paper's distinction between online publication, which requires connectivity, and local code creation, which does not. Internet shutdowns are a deliberate form of such outages. Zhang and Zhu (2011) find that blocking access to Chinese Wikipedia reduced contributions even among contributors who remained unblocked, consistent with a loss of social benefits from participation. Aceto et al. (2018) provide a comprehensive survey of Internet outages, and Schachenhofer et al. (2023) analyze how Internet blackouts affect information flows in organizations. Tagat, Phokeer, and Kreitem (2024) develop an econometric framework for measuring shutdown costs across countries and emphasize impacts on output, employment, and investment, and Gunsilius and Van Dijcke (2023) estimate reductions in economic activity from a regional Internet shutdown in India using a discontinuity-based method. These studies motivate the hypothesis that intentional connectivity losses can generate immediate real economic costs.

Measuring software production with GitHub data. A third literature uses digital trace data to measure software work. McDermott and Hansen (2021) use real-time GitHub activity to document a shift toward weekend and out-of-hours work during the COVID-19 pandemic. Nagle (2019) links the use of open-source software to firm productivity, finding positive value-added returns for firms with complementary capabilities, which helps establish that open-source activity is economically meaningful production. GitHub is especially useful because public events are timestamped and tied to concrete production and collaboration tasks. Wachs et al. (2022) use GitHub to map the geography of open-source software production, and Birkholz and Gomtsyan (2026) use GitHub

collaboration data to construct a global network of software production across locations and to study software development as tradable services. Hu et al. (2023) study how pandemic lockdowns affected open-source contributions in China and show that the direction of the effect can depend on context and adaptation. Kreitmeir and Raschky (2024) compile daily GitHub outcomes to study Italy's temporary ban on ChatGPT. Their design is particularly close methodologically: because GitHub actions are relatively rare at the user-day level, they transform count outcomes into binary daily indicators and report continuous outcomes as a robustness check. I follow that logic by making the daily active-user rate the primary outcome and treating counts as secondary outcomes.

Comparative-case causal inference. Synthetic control methods construct a weighted combination of untreated units that approximates the treated unit before an intervention. The approach originates in Abadie and Gardeazabal (2003), who estimate the economic costs of terrorist conflict in the Basque Country relative to a synthetic comparison region, an early application to a major political disruption, and was formalized by Abadie, Diamond, and Hainmueller (2010). Abadie (2021) emphasizes transparent donor selection, pre-treatment fit, and placebo or leave-one-out exercises as central diagnostics. Synthetic difference-in-differences combines reweighting of units and time periods with difference-in-differences structure (Arkhangelsky et al., 2021). I use conventional synthetic control as the headline specification because its weights and pre-fit are transparent in this application. Finally, I use regression discontinuity in time as an additional temporal check, following the cautions in Hausman and Rapson (2018) regarding serial correlation, functional form, and the conceptual differences between time-based and cross-sectional RD designs.

4 Data and Measurement

4.1 GitHub users and country assignment

The analysis begins from GitHub accounts returned by country-location searches for Bangladesh and fourteen potential donor countries: the Philippines, Pakistan, Sri Lanka, Nepal, Indonesia, Vietnam, Malaysia, Thailand, Kenya, Nigeria, Ghana, Morocco, Jordan, and Tunisia. GitHub profile location is self-reported. I therefore interpret the sample as accounts associated with each country rather than verified physical residence at every point in time.

For each account, I use GH Archive to recover public daily GitHub events from May 1 through September 4, 2024. Eligibility counts days with any recorded event, including days with zero `productive_events`, from May 1 through July 17, and requires at least four such days. All event dates are UTC. The stored selection also excludes suspected automation accounts, but the original login-pattern list and behavioral thresholds are unavailable. Reconstructing the activity threshold

yields 60,729 candidates, versus 59,158 retained accounts; the reason for each of the 1,571 exclusions cannot be independently reconstructed. Both stored bot flags are false for every retained account, which verifies their retained status but does not recover the exclusion rule. The final eligible sample contains 59,158 users, including 10,241 accounts associated with Bangladesh. This restriction focuses the design on developers with observable pre-shutdown activity and reduces the influence of dormant accounts. It also means that the estimand is an effect on pre-active public GitHub users, not on every person who has a GitHub account.

4.2 Outcomes

The primary outcome is the daily active-user rate: the share of eligible users in a country who record at least one productive GitHub action on day t. Operationally the indicator is $1\{\texttt{productive_events} > 0\}$. The exact GH Archive event-type mapping is not recoverable from the local aggregates: 104,786 of 1,103,680 eligible active user-days have no recorded push, pull-request opening/closure, issue opening/closure, release, review, or comment. The upstream mapping is therefore an unresolved measurement limitation, not an assumed sum of those components. This extensive-margin measure is preferred because GitHub activity is sparse at the user-day level and raw commit counts contain extreme outliers. Binary activity measures are also closely aligned with the approach used by Kreitmeir and Raschky (2024).

I construct secondary outcomes along the software-production pipeline. The secondary productive-output indicator indicates a day with commits, issue closure, pull-request closure, or a release. Shipping activity indicates a day with a merged pull request or a release. Collaboration activity indicates a day with code review or comment activity. I also construct count outcomes per 1,000 eligible users: collaboration events and shipping output. These measures separate recorded output from downstream delivery and coordination. The primary and secondary productive indicators differ: their Bangladesh pre-period means are 17.25 and 15.38 percent, respectively. Table 2 distinguishes them explicitly; neither outcome nor its estimates have been silently redefined.

A small number of accounts generate implausibly large daily or weekly commit counts, so the primary country-day synthetic-control analysis does not rely on a raw commit-count headline. National holidays are coded at the country-day level and used in robustness checks.

4.3 Commit-level mechanism data

To distinguish coding from publication, I build a separate commit-level dataset for Bangladesh. I extract public PushEvents by eligible Bangladeshi users, retain PushEvents for which the historical payload reveals the complete set of commit SHAs, and keep the earliest observed push for each repository-SHA pair. I then query GitHub commit metadata to recover `authoredDate`,

Table 2: Data and outcome construction

Component	Definition / sample	Scale
Primary outcome	Stored `productive_events>` 0; exact event-type map unresolved	Country-day percentage points
Secondary productive output	Any commits, issues closed, pull requests closed, or releases	User-day indicator / country-day rate
Shipping activity	Any merged pull request or release	User-day indicator / country-day rate
Collaboration activity	Any PR review, issue comment, PR review comment, or commit comment	User-day indicator / country-day rate
Collaboration events	Total review/comment events	Per 1,000 eligible users
Shipping output	Merged pull requests plus releases	Per 1,000 eligible users
Mechanism sample	Public repo-SHA pairs from fully observed PushEvents, matched to commit metadata	Commit level

Notes: Public GitHub events do not measure private repository activity, work on other platforms, meetings, or local code that is never pushed publicly.

`committedDate`, and GitHub-linked author and committer logins. The metadata-query target contains 86,065 repository-SHA pairs, rather than the full pre-period universe: only 17,000 of 134,479 pairs first pushed before July 18 (12.64 percent) were queried, while all 69,065 pairs first pushed on or after July 18 were targeted; 70,576 commit objects are successfully resolved, an 82.0 percent resolution rate.

The strict mechanism sample requires (i) a non-missing author timestamp, (ii) a nonnegative author-to-push delay, and (iii) equality between the GitHub login linked to the commit author and the PushEvent actor. This leaves 59,920 commits. The equality condition avoids attributing a commit written by another person to the Bangladeshi user who happened to push it. Git author and committer timestamps are embedded metadata rather than independently verified work-time measurements, so they are interpreted as evidence consistent with offline production rather than as an infallible clock of when coding occurred.

4.4 Sector-recovery sample and repository classification

To study the period after connectivity returned, I construct a separate within-Bangladesh panel. It starts from the 23,866 Bangladesh-associated accounts with at least one day of selected public GitHub activity between January 18 and July 17, 2024, and draws a fixed sample of 20,000 of them. Eligibility depends only on pre-shutdown activity, so the sample composition cannot respond to the shutdown or to post-shutdown behavior. For these accounts, I recover all public GH Archive events from January 18 through November 23, 2024, aggregated to the user-repository-day level. This sample differs from the 59,158-account synthetic-control sample in its eligibility rule (one rather

than four pre-treatment active days), its observation window, and its restriction to Bangladesh, so its activity levels are not directly comparable with the country-day rates in Section 6.

Each repository is assigned to a two-digit sector of the North American Industry Classification System (NAICS) by a publicly released multilingual transformer classifier that predicts a NAICS sector from repository text (`aquiro1994/naics-github-classifier-multilingual`; Quispe and Xu, 2026). The classifier input is the repository name and a historical snapshot of the root README file. For repositories observed on or before July 17, 2024, the snapshot is taken as of July 17; for repositories first observed after the shutdown, it is taken as of their first observed activity date. Current repository descriptions and topics are deliberately excluded, because they may have been written after the study period. Of the 168,875 distinct repositories in the panel, a historical README is recovered for 92,979 (55.1 percent); the remainder lack a root README at the target date or could not be retrieved. The main specification assigns a sector only when the classifier's confidence score is at least 0.80, which holds for 32,491 repositories across all sectors; repositories below this threshold do not contribute to any sector. Thresholds of 0.70 and 0.90 are used as robustness checks.

The sector measures the economic sector associated with the repositories in which observable GitHub activity occurs. It is not the developer's employer or industry of employment: the same developer may contribute to repositories classified in several sectors, and a repository's sector describes the application domain inferred from its text rather than the organization that funds the work.

5 Empirical Strategy

5.1 Synthetic control

The headline design treats Bangladesh as the single treated country and the fourteen other countries as the donor pool. Let Y_{ct} denote the daily active-user rate for country c on date t. Synthetic-control weights are nonnegative, sum to one, and are chosen to minimize the mean squared difference between Bangladesh and the weighted donor combination over the pre-treatment period May 1–July 17. The estimated daily treatment effect is the gap between Bangladesh and the synthetic control during July 18–23.

The method is well suited to the setting because treatment occurs at the country level and the number of aggregate comparison units is modest. Its main advantage here is transparency: the counterfactual is an explicit weighted combination of observed countries, and the credibility of the design can be evaluated through pre-treatment fit, donor weights, in-space placebos, and leave-one-

out exercises (Abadie, Diamond, and Hainmueller, 2010; Abadie, 2021).

$$\mathrm{ATT}_{\mathrm{SC}} = \frac{1}{|\mathcal{T}_1|} \sum_{t \in \mathcal{T}_1} \Big[Y_{\mathrm{BD},t} - \sum_j w_j Y_{j,t} \Big], \tag{1}$$

where $\mathcal{T}_1$ denotes the shutdown days.

5.2 Difference-in-differences and event study

As a complementary design, I compare the change in Bangladesh's mean outcome from the pre-period to the shutdown with the average corresponding change across the donor countries:

$$\mathrm{ATT}_{\mathrm{DiD}} = \big[\bar{Y}_{\mathrm{BD},1} - \bar{Y}_{\mathrm{BD},0}\big] - \frac{1}{14} \sum_{j=1}^{14} \big[\bar{Y}_{j,1} - \bar{Y}_{j,0}\big], \tag{2}$$

where $\bar{Y}_{c,0}$ is country c's mean daily active-user rate over May 1–July 17 and $\bar{Y}_{c,1}$ its mean over July 18–23. Unlike the synthetic control, which chooses donor weights to reproduce Bangladesh before the shutdown, this comparison gives equal weight to every donor. It is used primarily as a magnitude check against the synthetic-control estimate.

To trace the dynamics of the gap, I replace the shutdown window with weekly event periods k and compute

$$\hat{\theta}_k = \big[\bar{Y}_{\mathrm{BD},k} - \bar{Y}_{\mathrm{BD},0}\big] - \frac{1}{14} \sum_{j=1}^{14} \big[\bar{Y}_{j,k} - \bar{Y}_{j,0}\big], \tag{3}$$

where $\bar{Y}_{c,k}$ is country c's mean daily active-user rate in event period k; $k = -11, \ldots, -1$ are seven-day bins ending July 17 (bin -11 also contains May 1), $k = 0$ is the shutdown (July 18–23), and $k = +1, \ldots, +6$ are seven-day bins from July 24 to September 3. Equation (3) is computed directly from the stored country-day panel of the fifteen countries; it involves no regression and no user-level fixed effects. By construction, $\hat{\theta}_0 = \mathrm{ATT}_{\mathrm{DiD}}$, so the shutdown-week coefficient reproduces the aggregate DiD magnitude. Under parallel pre-shutdown paths, the coefficients for $k < 0$ would fluctuate narrowly around zero. As shown in Section 6.2, Bangladesh's pre-period is volatile and does not display a clean conventional parallel-trends pattern. The weekly coefficients are therefore not interpreted as an identified dynamic causal treatment path. They are diagnostic evidence on whether the largest break in the series is concentrated at the shutdown, and they complement, rather than replace, the synthetic control.

Because treatment varies only at the country level and there are fifteen countries, I do not rely on user-clustered standard errors. Uncertainty is instead summarized with in-space placebos: each donor is assigned the treatment in turn and compared with the equal-weighted mean of the remaining

thirteen donors. I report the standard deviation of the fourteen placebo estimates, a donor-placebo reference interval equal to $\hat{\theta}_k \pm 1.96$ times that standard deviation, and the rank of Bangladesh's absolute estimate among the fifteen units. These reference intervals describe how far donor countries move under the same calculation; they are not conventional sampling confidence intervals, and because Bangladesh's pre-period is more volatile than any donor's, they are not fully calibrated measures of uncertainty for Bangladesh.

5.3 Regression discontinuity in time

A regression discontinuity in time provides a third temporal comparison. The archived preferred specification uses a cutoff at July 17, 2024, a bandwidth of $h = 7$ days (July 10–24, fifteen daily observations), and a local linear fit on each side. With $\tilde{t} = t - \text{July 17}$ and $D_t = \mathbf{1}\{\tilde{t} \geq 0\}$, so that July 17 itself lies on the post-cutoff side, I estimate

$$Y_t = \alpha + \tau D_t + \gamma_1 \tilde{t} + \gamma_2 D_t \tilde{t} + u_t, \qquad |\tilde{t}| \leq h, \tag{4}$$

where τ is the discontinuity and Y_t is the daily share of users with at least one productive GitHub event in a balanced user-day panel of 10,329 Bangladesh-associated and 3,025 Philippines-associated accounts. This eligible sample differs slightly from the 10,241 Bangladesh accounts of the country-day panel used for the synthetic control. I estimate Equation (4) for Bangladesh, for the Bangladesh-minus-Philippines gap as a controlled specification, and for the Philippines alone as a geographic placebo. Inference uses heteroskedasticity-and-autocorrelation-consistent (HAC) Newey–West standard errors with two lags, following the cautions in Hausman and Rapson (2018) regarding serial correlation in time-series discontinuity designs. Because time itself is the running variable and other contemporaneous events can occur at the same date, the RDiT is interpreted as a robustness check rather than a standalone source of causal identification.

5.4 Inference and robustness

For synthetic control, I implement in-space placebos by assigning the treatment date to each donor country in turn and reconstructing its synthetic control using the remaining donors. The primary rank statistic compares Bangladesh's absolute ATT and post/pre RMSPE ratio with the placebo distribution. Abadie, Diamond, and Hainmueller (2015) illustrate the use of post/pre RMSPE ratios to scale post-intervention discrepancies by pre-intervention fit, alongside in-time placebos that reassign treatment to earlier dates. With fourteen placebo countries, the smallest possible finite-sample rank value is $1/(14 + 1) = 0.0667$; consequently, a value of 0.0667 means Bangladesh is the most extreme unit, not that the effect is weak.

I also re-estimate the primary synthetic control after removing each donor country one at a time. Holidays are excluded in a separate DiD specification. Finally, outcome-specific synthetic controls allow donor weights to vary by outcome, whereas the main secondary-outcome table deliberately reuses the active-rate synthetic-control weights. Reusing the primary weights reduces the risk that each new outcome is matched with a different donor combination chosen after observing results.

5.5 Within-Bangladesh sectoral recovery

The designs above estimate the effect of the shutdown during July 18–23. To describe how activity evolved after restoration, I use the fixed sample of 20,000 accounts described in Section 4.4. For user i, sector s, and day t, let $A_{ist} = 1$ if the user records at least one PushEvent on day t in a repository classified to sector s; activity in several repositories of the same sector on the same day counts once. Days are grouped into event periods k: the final eight complete pre-shutdown weeks, $k = -8, \ldots, -1$ (May 23–July 17, 2024); the shutdown, $k = 0$ (July 18–23); and consecutive seven-day bins after it, starting with $k = +1$ (July 24–30). Let $\bar{A}_{isk}$ be user i's share of active days in period k, and $\bar{A}_{is}^{\text{pre}}$ the corresponding share over the 56 baseline days. I report

$$\beta_{sk} = \frac{1}{N}\sum_{i=1}^{N}\left(\bar{A}_{isk} - \bar{A}_{is}^{\text{pre}}\right), \qquad R_{sk} = 100 \times \frac{\sum_i \bar{A}_{isk}}{\sum_i \bar{A}_{is}^{\text{pre}}}, \tag{5}$$

where $N = 20{,}000$ and users without activity in sector s contribute zeros. The coefficient β_{sk} is the change in the daily probability of sector-s push activity relative to the baseline, in percentage points; the recovery index R_{sk} normalizes the baseline to 100. Standard errors for β_{sk} are the standard deviation of the user-level differences divided by $\sqrt{N}$, and 95 percent intervals use the normal approximation. As a summary of recovery timing, I report the first post-shutdown week that begins two consecutive weeks with $R_{sk} \geq 80$.

Because the activity measure counts pushes to any classified repository, including repositories first observed after the shutdown, it captures recovery through new projects and through switching across repositories. As a complementary specification, I also follow only the repositories pushed to during the baseline weeks, whose activity cannot be replaced by new projects.

This analysis compares each sector with its own pre-shutdown baseline and has no untreated comparison group. It therefore describes the heterogeneity and timing of recovery within Bangladesh; it does not estimate sector-specific causal effects. Post-shutdown deviations from the baseline may also reflect trends, seasonality, or other Bangladesh-specific developments. The confidence intervals capture sampling variation across users but not common shocks that affect all users in a given week. The pre-shutdown coefficients ($k < 0$) indicate how much weekly activity fluctuates around the baseline in the absence of a shutdown and serve as a benchmark for

interpreting post-shutdown deviations.

For the RDiT, July 17 is retained to make the historical estimate reproducible, not because its effect is preferable. Provider-level mobile outages began late on July 17 UTC, but the nationwide near-total cutoff was around 15:00 UTC July 18 (Belson, 2024a). Neither a July 17 midnight nor a July 18 midnight cutoff exactly identifies that event. A complete July 17/18/19 by 4/5/7/10/14-day grid is reported in Table A8, with the separate archived 16-specification grid in Table A9. A change to the preferred cutoff or an hourly design requires a separate methodological decision.

6 Results

Table 3 summarizes the estimates for the primary outcome from the three designs. The synthetic control is the primary causal specification; the international difference-in-differences and the regression discontinuity in time are complementary analyses of the same shock. The subsections below present each design's main figure and table.

Table 3: Summary of estimates across designs: daily active-user rate

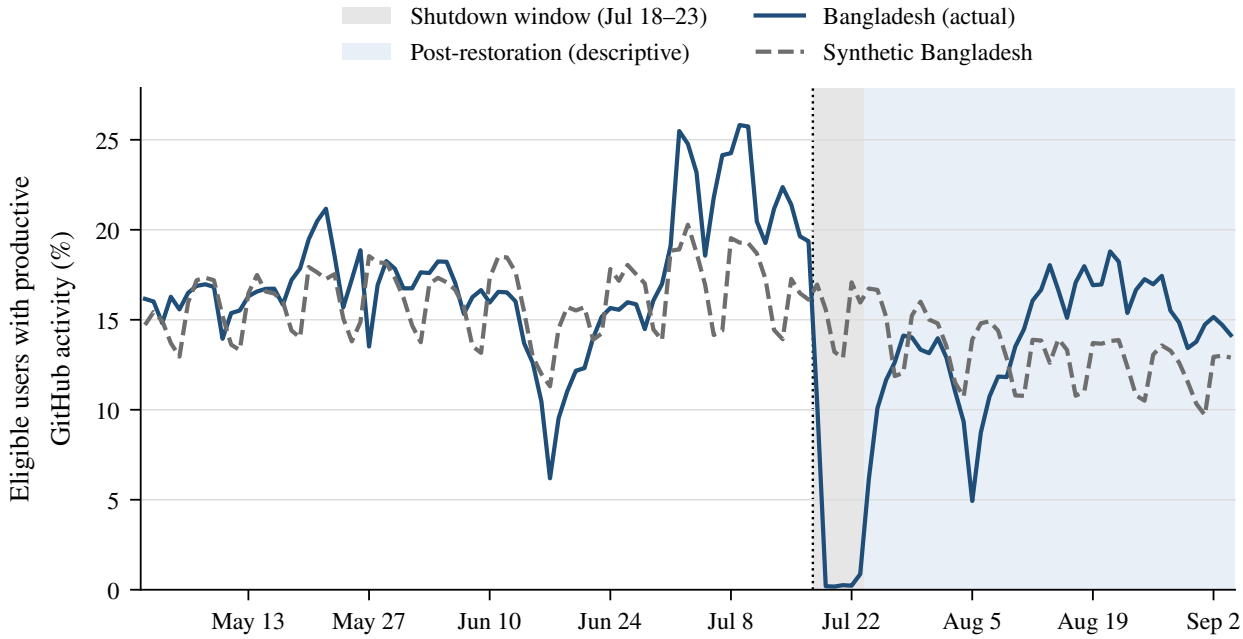
Design	Effect (pp)	Relative effect	Inference	Role
Synthetic control	−13.256	−86.8%	In-space placebo rank: 1/15 = 0.0667 (14 placebo countries)	Primary causal design
International DiD	−13.088	−86.6%	Donor-placebo rank: 1/15; placebo SD 1.17 pp	Magnitude check
DiD, holidays excluded	−13.379	−86.9%	Donor-placebo rank: 1/15; placebo SD 1.19 pp	Robustness
RDiT, local linear	−9.284	–	HAC $p = 0.065$ (SE 5.03 pp)	Temporal robustness

Notes: Outcome: daily percentage of eligible users with observable productive GitHub activity; effects are averages over July 18–23, 2024, in percentage points. The inference column reports different objects that are not directly comparable. The rank values are the share of the 15 countries whose estimate is at least as extreme as Bangladesh's; 1/15 is the minimum attainable and means that Bangladesh is the most extreme unit, not $p < 0.05$. The RDiT p-value is a conventional time-series (HAC) p-value for the preferred specification (cutoff July 17, bandwidth 7 days, degree 1). Levels, fit statistics, and full inference are in Tables 4 to 6.

6.1 Synthetic control: main effect on daily productive activity

The synthetic-control estimate shows an exceptionally large collapse in observable GitHub activity (Figure 1 and Table 4). Bangladesh's active-user rate averages 2.021 percent during July 18–23, compared with a synthetic counterfactual of 15.277 percent. The ATT is −13.256 percentage points, equivalent to an 86.8 percent reduction relative to the synthetic counterfactual. The pre-treatment RMSPE is 3.260 percentage points and the post/pre RMSPE ratio is 4.192.

Figure 1: Actual and synthetic Bangladesh: daily share of eligible users with observable productive GitHub activity



Notes: Daily percentage of eligible Bangladesh-associated accounts with observable productive GitHub activity (solid) and its synthetic control (dashed; Pakistan 47.6%, Ghana 32.5%, Kenya 19.9%, weights fit on May 1–July 17; Table 4). The shaded shutdown interval defines the primary treatment estimand. Post-restoration differences are shown descriptively and are not interpreted as additional causal treatment effects. The blue band (July 24–September 4) extends the synthetic series with the same weights; average gaps in weeks +4 and +5 (4.24 and 4.67 percentage points) remain within the range of pre-treatment weekly gaps (−2.86 to 5.88). June 17 coincides with a national holiday. August 4–5 includes a second Internet shutdown; August 5 also falls in an initially declared holiday period, so the dip cannot be attributed to a holiday alone (Belson, 2024c; Bangladesh Sangbad Sangstha, 2024). Pre-treatment fit is imperfect (RMSPE 3.26 percentage points), most visibly in early July. Source: Author's calculations using GH Archive.

Keeping the original donor weights, the complete low-connectivity days July 19–22 yield actual activity of 0.212 percent, a 14.687 percent synthetic counterfactual, and a gap of −14.475 percentage points (−98.55 percent), alongside the primary July 18–23 estimate of −13.256 percentage points (−86.77 percent). The narrower window is a reported sensitivity, not a replacement for the primary estimand. Its residual activity averages 21.75 accounts per day. If every such active account were outside Bangladesh, location error among those observed active would account for at most 0.212 percent of the eligible cohort per day. This is not an upper bound on all incorrectly located accounts: inactive accounts' physical locations remain unobserved, and exceptional connectivity could also explain residual activity.

The synthetic control places 47.6 percent weight on Pakistan, 32.5 percent on Ghana, and 19.9 percent on Kenya (Table 4). No other donor receives positive weight in the baseline optimization. After restoration, shown descriptively in Figure 1, the average weekly gap between Bangladesh and its synthetic counterpart narrows to about −3 percentage points in the first two weeks (−3.11 and

Table 4: Synthetic control: main estimates and placebo inference

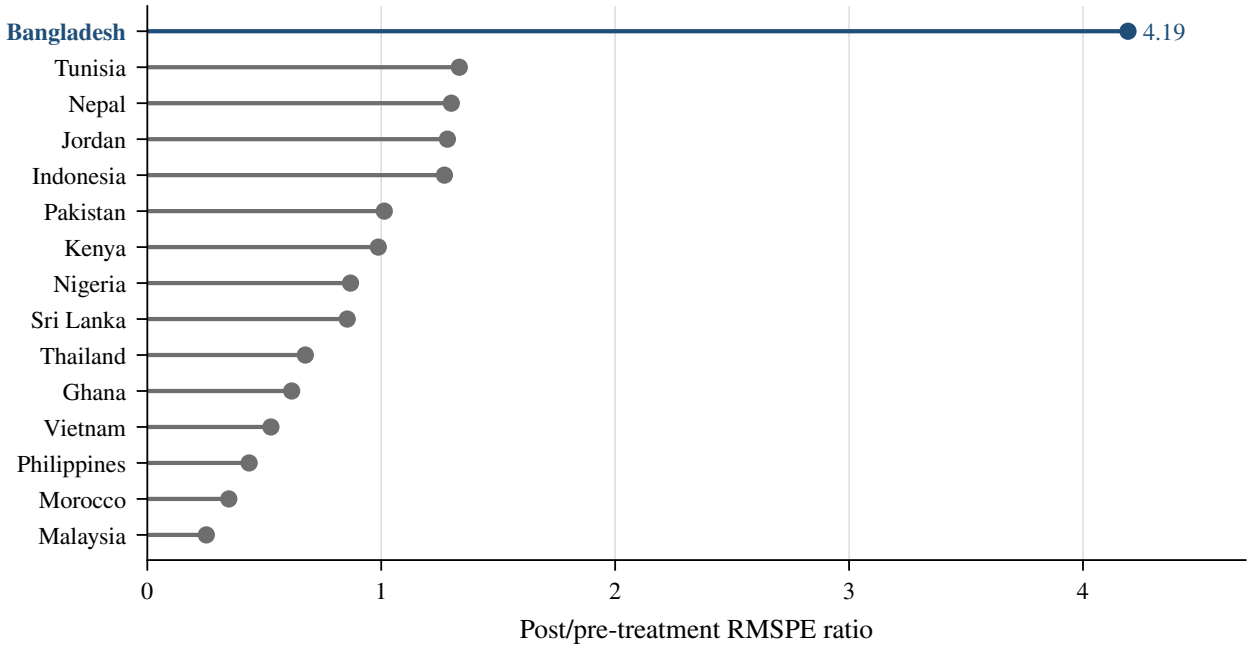
Statistic	Value
A. Estimates, July 18–23, 2024	
Bangladesh, actual (%)	2.021
Synthetic counterfactual (%)	15.277
ATT (percentage points)	−13.256
Relative effect	−86.8%
B. Fit and in-space placebo inference	
Pre-treatment RMSPE (pp)	3.260
Post-treatment RMSPE (pp)	13.666
Post/pre RMSPE ratio	4.192
Placebo countries	14
Rank p-value, absolute ATT	0.0667
Rank p-value, RMSPE ratio	0.0667
C. Donor weights	
Pakistan	47.6%
Ghana	32.5%
Kenya	19.9%

Notes: Outcome: daily percentage of eligible users with observable productive GitHub activity. Weights are chosen on May 1–July 17, 2024; the other eleven donors receive zero weight. Rank p-values compare Bangladesh with 14 in-space placebos; $1/15 = 0.0667$ is the minimum attainable value, so it indicates that Bangladesh is the most extreme unit, not $p < 0.05$. Full placebo ranking in Table A2. Sources: `results/synthetic_control/sc_inspace_placebos.csv`, `results/final/table_primary_SC_weights.csv`.

−2.89 percentage points) and turns positive in the third week; the positive average gaps in weeks +4 and +5 (4.24 and 4.67 percentage points) lie within the range of pre-treatment weekly gaps and are not interpreted as effects of the shutdown or of restoration. Bangladesh is more extreme than all fourteen in-space placebos in both absolute ATT and RMSPE-ratio comparisons, yielding rank statistics of 0.0667, the smallest value attainable with fourteen placebo units (Figure 2 and Table A2).

The leave-one-out exercise is highly stable (Table 10 in Section 8.1). Across specifications that remove one donor at a time, the ATT ranges only from −13.256 to −12.886 percentage points. In particular, excluding Pakistan produces −12.905 percentage points, excluding Ghana produces −13.209 percentage points, and excluding Kenya produces −12.886 percentage points. This matters because both Pakistan and Kenya experienced localized connectivity disruptions near the broader pre-treatment period: Internet Society reports regional restrictions in Pakistan around July 14–17 (Internet Society, 2024b,c), while Cloudflare documents a major Kenyan traffic disruption on June 25 (Belson, 2024b). The limited movement in the treatment estimate when either donor is removed reduces, though does not eliminate, concern that these donor-specific episodes drive the result.

Figure 2: Synthetic-control in-space placebo inference: post/pre-treatment RMSPE ratios



Notes: Each donor is assigned the July 18–23 treatment in turn, with its synthetic control rebuilt from the remaining donors. Bangladesh has the largest post/pre RMSPE ratio and the largest absolute ATT of the 15 units, so the exact rank p-value is $1/15 = 0.0667$ for both, the minimum attainable with 14 placebo countries. Bangladesh is the most extreme unit, but the resolution of this inference is limited by the size of the donor pool. Source: Author's calculations using GH Archive.

6.2 Difference-in-differences and event study

The international DiD estimate is −13.088 percentage points, nearly identical to the synthetic-control ATT (Table 5, Panel A). Removing national holidays yields −13.379 percentage points. The agreement is notable because the two methods use different weighting logic: synthetic control chooses donor weights to reproduce Bangladesh before the shutdown, whereas the international DiD gives equal weight to donor-country changes. Bangladesh's estimate is also more extreme than that of every in-space placebo, whose estimates range from −1.72 to 1.60 percentage points; as for the synthetic control, the resulting rank value of 0.0667 is the minimum attainable with fifteen countries.

Cumulative shortfall and persistence. Using the unchanged synthetic weights and 10,241-account cohort, the July 18–23 gap sums to 8,145 active user-days. The first daily return to the synthetic level is July 27, by which point the cumulative deficit is 10,186 user-days. A daily crossing is not sustained recovery: the first nonnegative trailing seven-day mean gap is August 13 (cumulative deficit 12,068); the first seven consecutive nonnegative daily gaps end August 16. Returning to Bangladesh's own pre-period mean is a different criterion: the first daily crossing is

August 14 and the first seven-day mean crossing August 22.

Through September 4, negative daily gaps total 13,526 user-days and positive gaps total 9,893, leaving a net deficit of 3,633. The cumulative deficit peaks at 13,237 on August 9 and is never fully offset within observation. Positive post-restoration gaps indicate an arithmetic offset, but cannot establish causal catch-up: early-July pre-treatment elevation, later domestic events, and donor disturbances also affect this counterfactual. Fifteen of 43 post-window days remain below it. These are descriptive sums of the original gap, not losses in hours worked or total software production; persistence after September 4 is not observed in this national panel.

Table 5: International difference-in-differences and aggregate event study

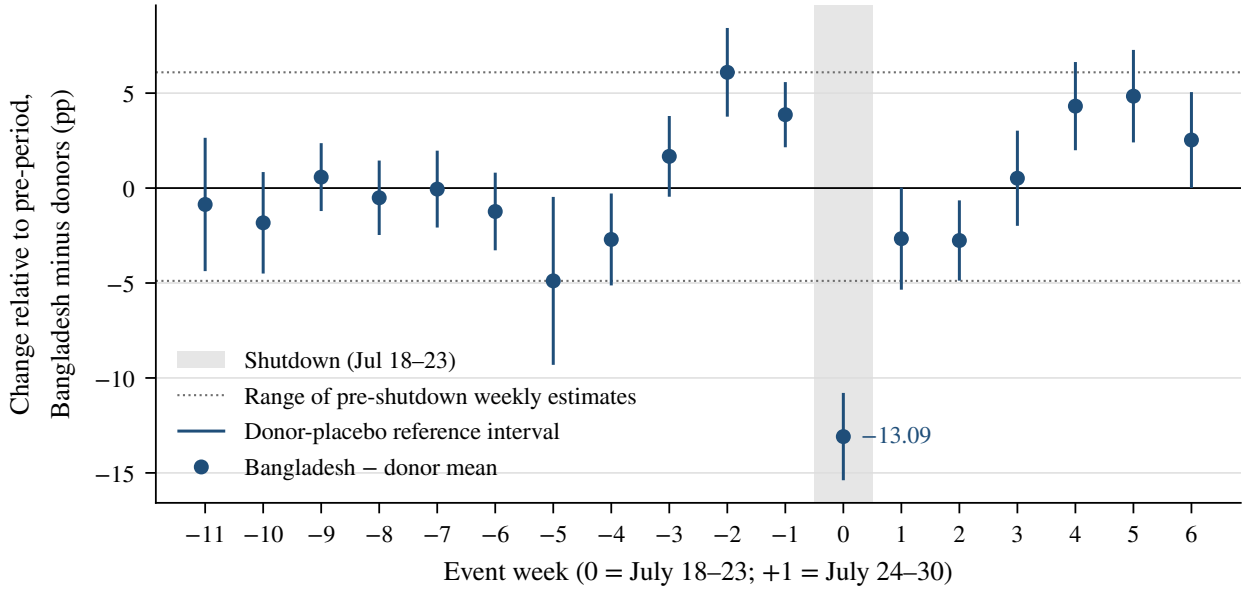
Specification	BD pre (%)	BD shutdown (%)	Donor change	ATT (pp)	Relative effect	Placebo SD	Rank p
A. Difference-in-differences, July 18–23 vs. May 1–July 17							
All days	17.25	2.02	−2.15	−13.09	−86.6%	1.17	0.0667
Holidays excluded	17.58	2.02	−2.18	−13.38	−86.9%	1.19	0.0667
B. Event study, selected weeks (all weeks in Table A6)							
Event week (dates)				$\hat{\theta}_k$ (pp)	Placebo-based interval	Placebo SD	Rank p
−5 (Jun 13–Jun 19)				−4.89	[−9.31, −0.47]	2.26	0.0667
−2 (Jul 4–Jul 10)				6.10	[3.76, 8.43]	1.19	0.0667
−1 (Jul 11–Jul 17)				3.86	[2.15, 5.58]	0.88	0.0667
0 (Jul 18–Jul 23)				−13.09	[−15.39, −10.79]	1.17	0.0667
+1 (Jul 24–Jul 30)				−2.66	[−5.35, 0.02]	1.37	0.0667
+2 (Jul 31–Aug 6)				−2.76	[−4.87, −0.65]	1.08	0.0667
+3 (Aug 7–Aug 13)				0.52	[−1.99, 3.02]	1.28	0.9333
+4 (Aug 14–Aug 20)				4.31	[1.99, 6.64]	1.18	0.0667
+5 (Aug 21–Aug 27)				4.84	[2.40, 7.28]	1.24	0.0667
+6 (Aug 28–Sep 3)				2.53	[0.01, 5.05]	1.29	0.0667

Notes: Outcome: daily percentage of eligible users with observable productive GitHub activity. Panel A compares Bangladesh's change from May 1–July 17 to July 18–23 with the equal-weighted mean change of the 14 donor countries (Equation (2)); the ATT reproduces `results/final/table_DiD_all_outcomes.csv`. Panel B reports $\hat{\theta}_k$ (Equation (3)) for weekly bins; week 0 equals the Panel A all-days ATT. Weeks −5 and −2 are the most negative and most positive pre-shutdown estimates. Inference uses in-space placebos: each donor is compared in turn with the mean of the other 13. *Placebo SD*: standard deviation of the 14 placebo estimates. *Placebo-based interval*: $\hat{\theta}_k \pm 1.96\times$SD, a donor-placebo reference interval, not a conventional sampling confidence interval. *Rank p*: share of the 15 units whose absolute estimate is at least as large as Bangladesh's (minimum 0.0667). Bangladesh's pre-period weekly estimates are more volatile than any donor's, so these placebo-based measures are not fully calibrated sampling uncertainty for Bangladesh. The pre-period does not show a clean parallel-trends pattern, and the weekly coefficients are a diagnostic, not an identified dynamic treatment path; the synthetic control remains the main design. Sources: `results/did/did_main_placebo_inference.csv`, `results/did/did_eventstudy_weekly.csv`.

Figure 3 reports the weekly event study from Equation (3). Its shutdown-week coefficient, −13.09 percentage points, reproduces the aggregate DiD estimate and is more than twice as large in absolute value as any pre-shutdown weekly coefficient. The pre-shutdown period, however, is volatile and does not display a clean conventional parallel-trends pattern. Weekly coefficients range from −4.89 percentage points (week −5, June 13–19) to 6.10 (week −2, July 4–10), and Bangladesh

was above its pre-period position relative to the donors in the two weeks immediately before the shutdown. This mirrors the imperfect early-July fit of the synthetic control (Figure 1). Because the reference is the full pre-period mean, this elevated activity does not generate the shutdown estimate; measured against the immediately preceding week alone, the decline would be larger. Given this pre-period volatility, the weekly coefficients are not interpreted as an identified dynamic causal treatment path. They are diagnostic evidence that the largest break in the series is concentrated in the shutdown week.

Figure 3: Diagnostic event study: Bangladesh relative to the equal-weighted donor mean, by week



Notes: Points are $\hat{\theta}_k$ (Equation (3)): Bangladesh's change in the daily active-user rate relative to its May 1–July 17 mean, minus the equal-weighted mean change of the 14 donors, by seven-day bin. Week 0 is the July 18–23 shutdown (shaded) and equals the DiD estimate in Table 5. Vertical lines are donor-placebo reference intervals (±1.96 times the standard deviation of the 14 placebo estimates), not conventional confidence intervals, and are not fully calibrated because Bangladesh's pre-period is more volatile than any donor's. Dotted lines mark the range of Bangladesh's pre-shutdown coefficients. The pre-period shows no clean parallel-trends pattern; the coefficients are a diagnostic, not an identified dynamic treatment path. Source: Author's calculations using GH Archive.

The donor-placebo reference intervals in Figure 3 should be read with care. They are not conventional sampling confidence intervals: they scale the spread of the fourteen donor placebo estimates in each week. Bangladesh's pre-shutdown weekly coefficients have a standard deviation of 3.04 percentage points, compared with 0.70 to 2.18 percentage points for the donors treated as placebo units. Several pre-period coefficients for Bangladesh are therefore as extreme as any placebo, and the reference intervals are not fully calibrated measures of uncertainty for Bangladesh. The more informative comparison is with Bangladesh's own pre-shutdown range: the shutdown coefficient lies far outside it, whereas every post-restoration coefficient lies within it.

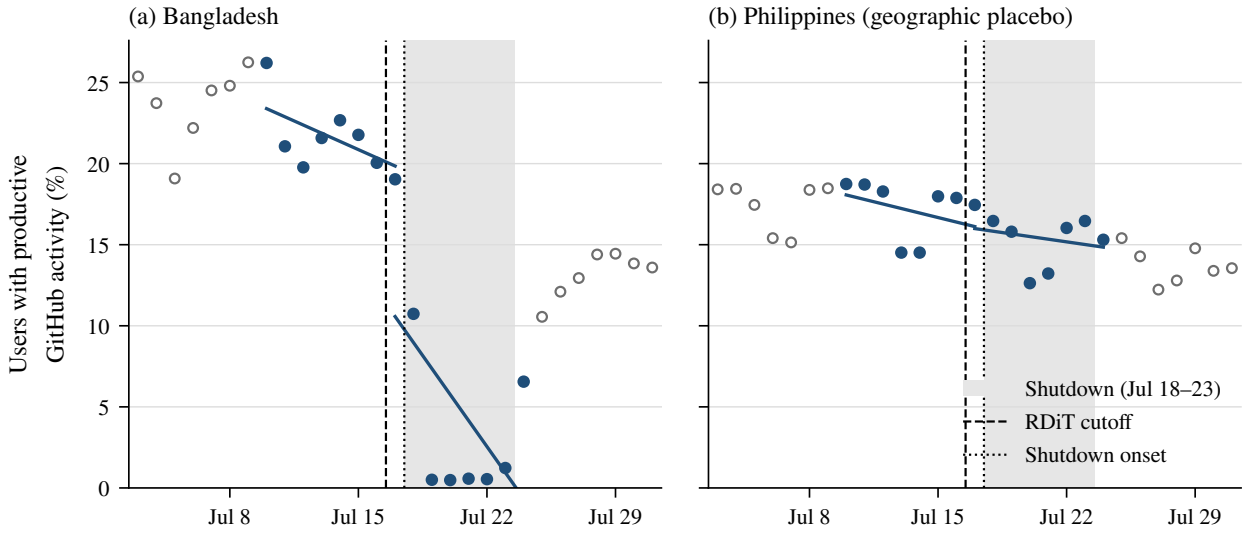
After restoration, the gap narrows quickly. The coefficients are −2.66 and −2.76 percentage

points in the two weeks after the shutdown and are positive from week +3 onward. The positive coefficients in weeks +4 to +6 (2.53 to 4.84 percentage points) lie within the range of pre-shutdown fluctuations and should not be interpreted as gains caused by the shutdown. Recovery dynamics within Bangladesh are examined by repository sector in Section 6.5.

6.3 Regression discontinuity in time

Figure 4 shows the daily series used in the RDiT. In Bangladesh, the share of users with productive activity is 19.03 percent on July 17, falls to 10.74 percent on July 18, and lies between 0.48 and 1.23 percent from July 19 through July 23 before recovering gradually from July 24. The Philippines, used as a geographic placebo, shows no comparable break.

Figure 4: Regression discontinuity in time: daily share of users with productive activity around the July 17 cutoff (visual companion to Table 6)



Notes: Daily percentage of users with productive GitHub activity in the balanced user-day panels used for the RDiT (10,329 Bangladesh-associated and 3,025 Philippines-associated accounts). Filled markers are the fifteen days in the 7-day bandwidth of Equation (4). Lines are the fits of the archived preferred specification and correspond exactly to Table 6. Dashed line: cutoff, with July 17 on the post-cutoff side; dotted line: shutdown boundary between July 17 and 18; shaded: July 18–23 shutdown. The RDiT is a temporal robustness check, not the primary design. Source: Author's calculations using GH Archive.

The RDiT produces a smaller point estimate but the same qualitative conclusion (Table 6). The preferred local linear specification estimates a discontinuity of −9.28 percentage points in Bangladesh's active-user rate; a controlled Bangladesh-minus-Philippines version yields −9.16 percentage points. HAC p-values are approximately 0.065 in both cases, while the Philippines placebo is near zero. The smaller magnitude than in the synthetic control and DiD is consistent with the design visible in Figure 4: the cutoff day, July 17, precedes the shutdown but lies on the post-cutoff side, and the linear post-cutoff fit spans both the collapse and the beginning of the recovery on July

24. Because Hausman and Rapson (2018) urge caution in interpreting RDiT estimates, I emphasize the direction and timing of the break rather than treating the RDiT p-value as the decisive inferential statistic.

Table 6: Regression discontinuity in time: preferred specification

Specification	Outcome	Effect (pp)	HAC SE	95% CI	p	Days
Main local linear	Bangladesh active rate	−9.284	5.034	[−19.151, 0.584]	0.065	15
Controlled local linear	Bangladesh – Philippines active rate	−9.160	4.980	[−18.921, 0.601]	0.066	15
Geographic placebo	Philippines active rate	−0.124	1.492	[−3.048, 2.801]	0.934	15

Notes: Local linear RDiT at the July 17, 2024 cutoff, bandwidth 7 days, polynomial degree 1, no donut, HAC lags 2. Effects are reported in percentage points. RDiT is a temporal robustness check, not the primary causal design. Source: `results/rdit/main_rdit_summary.csv`.

6.4 Productive, shipping, and collaboration outcomes

The collapse is not limited to a generic activity indicator. Holding the primary synthetic-control weights fixed, productive activity falls by 11.665 percentage points relative to a 13.469 percent counterfactual, or 86.6 percent. Shipping activity falls by 0.979 percentage points relative to a 1.049 percent counterfactual, or 93.3 percent. Collaboration activity falls by 0.611 percentage points relative to a 0.654 percent counterfactual, or 93.5 percent. Count outcomes show similarly large relative declines in collaboration events and shipping output (Table 7). International DiD estimates for the same outcomes, which point in the same direction, are reported in Table A7.

Allowing the donor weights to re-optimize separately for each outcome produces comparable magnitudes: −86.7 percent for productive activity, −90.5 percent for shipping activity, and −91.9 percent for collaboration activity. The in-space placebo ATT rank is again 0.0667 for all three key activity-rate outcomes. The RMSPE-ratio rank is 0.0667 for productive and shipping activity but 0.1333 for collaboration activity. Among the count outcomes, shipping output has a rank of 0.0667 on both statistics, whereas collaboration events have an ATT rank of 0.2667 and an RMSPE-ratio rank of 0.1333, so Bangladesh is not the most extreme unit for that outcome. I therefore treat collaboration as consistent secondary evidence rather than giving it the same inferential weight as the primary and shipping outcomes.

6.5 Sectoral recovery after restoration

The synthetic-control estimates describe the collapse during the shutdown window. They do not show how activity evolved once connectivity returned, or whether recovery was common across the kinds of projects on which Bangladeshi developers work. This subsection addresses these

Table 7: Secondary outcomes

	Fixed primary weights		Outcome-specific weights		Placebo rank	
Outcome	ATT	Relative	ATT	Relative	ATT	RMSPE ratio
Productive active rate	−11.665 pp	−86.6%	−11.720 pp	−86.7%	0.0667	0.0667
Shipping active rate	−0.979 pp	−93.3%	−0.668 pp	−90.5%	0.0667	0.0667
Collaboration active rate	−0.611 pp	−93.5%	−0.480 pp	−91.9%	0.0667	0.1333
Collaboration events / 1,000	−19.956	−95.5%	−11.640	−92.5%	0.2667	0.1333
Shipping output / 1,000	−17.913	−93.0%	−13.274	−90.8%	0.0667	0.0667

Notes: Fixed-weight specifications reuse the primary active-rate SCM weights (Table A1). Outcome-specific specifications re-estimate donor weights separately for each outcome (Table A3). Rank values are exact in-space placebo ranks for the outcome-specific specifications with 14 placebo countries; 0.0667 is the minimum attainable value. All outcomes are computed over eligible users in both the numerator and the denominator.

questions using the fixed sample of 20,000 Bangladesh-associated accounts and the repository-sector classification of Section 4.4, estimated as described in Section 5.5. The analysis compares each sector with its own pre-shutdown baseline. It documents heterogeneity in recovery dynamics within Bangladesh; it does not estimate sector-specific causal effects relative to an untreated international control, and the synthetic-control estimate in Section 6.1 remains the paper's headline result. Throughout, a sector refers to the economic sector associated with the repositories in which observable GitHub activity occurs, not to developers' employers or industries of employment.

Table 8: Recovery of observable GitHub push activity by repository sector

		Pre	Pre rate	Recovery index (pre-period mean = 100)					First two-
Sector	NAICS	users	(%)	Shutdown	+1	+2	+4	+8	week run ≥80
Finance	52	382	0.078	30.9	92.2	67.6	89.5	91.3	3
Information	51	1,528	0.442	11.7	75.9	72.2	106.0	94.5	3
Education	61	1,234	0.385	11.7	59.1	53.4	75.4	81.9	6
Health	62	514	0.132	6.3	40.7	51.5	77.5	89.4	6
Retail	44–45	1,012	0.295	6.2	52.0	52.3	153.7	79.6	9
Real Estate	53	213	0.063	4.0	33.0	25.1	44.4	85.5	5
Accommodation/Food	72	480	0.144	3.5	44.3	42.8	54.2	62.6	–

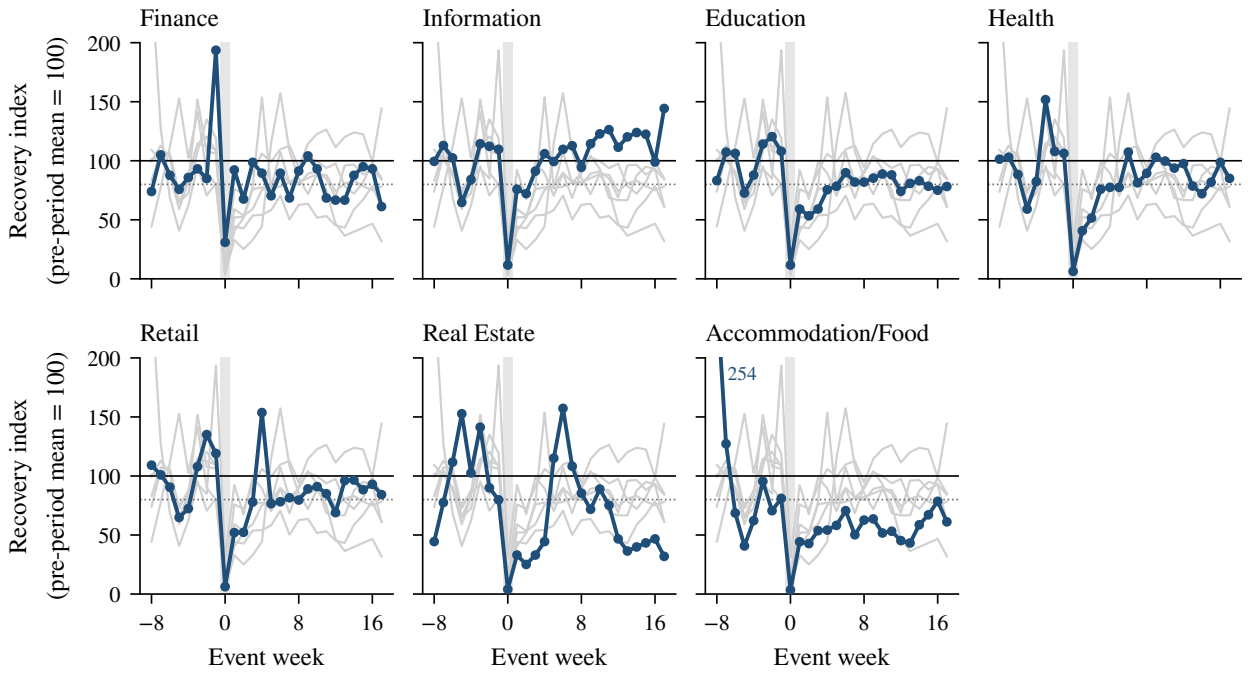
Notes: Fixed sample of 20,000 Bangladesh-associated accounts. Repository sectors from the historical-README classifier with confidence ≥ 0.80. The outcome is the share of sampled users with at least one PushEvent to a repository of the sector on a given day. *Pre users*: sampled users with such activity in the baseline period. *Pre rate*: mean daily outcome over the baseline period (May 23–July 17, 2024, the final eight complete pre-shutdown weeks), in percent of all 20,000 users. Shutdown: July 18–23; week +1: July 24–30; subsequent 7-day bins. *First two-week run ≥80*: first post-shutdown week that begins two consecutive weeks with an index of at least 80 (observed through week +17, November 13–19); it does not imply that the index stays above 80 thereafter. – indicates no such week. Within-Bangladesh comparison with the sector's own pre-period; not a causal effect relative to an untreated control. Sources: `results/sector_recovery/sectorwide/sectorwide_recovery_summary.csv`, verified against `results/sector_recovery/final/table_sector_recovery_main.csv`.

All seven sectors show a sharp fall in observable push activity during July 18–23 (Table 8 and Figure 5). The shutdown-period recovery index ranges from 3.5 in Accommodation/Food to

11.7 in Information and Education and 30.9 in Finance. In every sector, the 95 percent confidence interval for the change relative to the baseline excludes zero (Appendix Figure A2).

Restoration of connectivity did not imply an immediate or uniform recovery. In the first week after the shutdown (July 24–30), the recovery index ranges from 33.0 in Real Estate to 92.2 in Finance. Mobile connectivity returned only during this week, later than fixed broadband (Belson, 2024a), so part of the week +1 shortfall may reflect incomplete restoration. Recovery was also not monotonic: in Finance, Information, Education, Real Estate, and Accommodation/Food, the index is lower in week +2 (July 31–August 6) than in week +1. This week includes the renewed mobile shutdown on August 4 and the nationwide disruption on August 5 (Belson, 2024c), alongside unrest and initially declared holidays. These overlapping events prevent a separate attribution of the second-week decline. Although the stored holiday labels extend through August 7, offices were ordered to reopen on August 6 (Bangladesh Sangbad Sangstha, 2024); this conflict is preserved and audited rather than silently recoded.

Figure 5: Recovery of observable GitHub push activity by repository sector



Notes: Each panel highlights one sector's recovery index R_{sk} (Equation (5)), 100 times its mean daily probability of push activity in event week k relative to May 23–July 17, 2024; other sectors in grey. Shaded: July 18–23 shutdown; week +1 is July 24–30, and the window ends at week +17 (November 13–19). Lines mark 100 and 80; the one value above 200 is labeled. Fixed sample of 20,000 Bangladesh-associated accounts; sectors from a README classifier with confidence ≥ 0.80 (Quispe and Xu, 2026) describe the repositories in which activity occurs, not developers' employers. Within-Bangladesh descriptive comparison, not a causal effect. Source: Author's calculations using GH Archive.

Finance and Information recovered fastest. Finance returned to an index of 92.2 in week +1, and both sectors begin a run of two consecutive weeks at or above 80 in week +3. Information

exceeds its baseline by week +4 (106.0). In week +8, Finance is 91.3 and Information is 94.5. The Finance series is nonetheless noisy: its baseline includes a spike in week −1 (index 193.6), and its post-shutdown index falls back below 80 in several later weeks.

Education and Health recovered more gradually. Their week +1 indices are 59.1 and 40.7, they reach 75.4 and 77.5 by week +4, and both begin a run of two consecutive weeks at or above 80 only in week +6. By week +8, they stand at 81.9 and 89.4.

Accommodation/Food remained persistently depressed. Its index is 44.3 in week +1, 54.2 in week +4, and 62.6 in week +8, and it never records two consecutive weeks at or above 80 through the end of the observation window (week +17, November 13–19). This result should be read alongside the sector's unstable baseline: the index is 254.1 and 127.3 in the first two baseline weeks and between 40.8 and 95.5 in the remaining six. Part of the post-shutdown gap may therefore reflect a decline that began before the shutdown rather than a failure to recover.

Real Estate recovered slowly at first and later approached its baseline. Its index remains between 25.1 and 44.4 through week +4, rises above 100 in weeks +5, +6 and +7 (115.1, 157.3 and 108.3), and is 85.5 in week +8. This later recovery is not durable within the observed window: from week +12 onward, the index lies between 31.9 and 46.7 (Figure 5). Real Estate is also the smallest of the seven sectors, with 213 sampled users active in the baseline period, so its weekly estimates are imprecise.

Retail shows a pronounced temporary rebound. Its index is about 52 in weeks +1 and +2, rises to 153.7 in week +4 (August 14–20), and falls back to 76.5 in week +5. Its first run of two consecutive weeks at or above 80 begins only in week +9. The week +4 value is therefore a transient overshoot, possibly reflecting a burst of catch-up activity, rather than evidence of completed recovery; the data do not identify what generated it. The timing of Retail's first two-week run is also sensitive to the classifier threshold (Section 8.3).

Index values above 100 later in the window should not be read as gains caused by the shutdown. Information, for example, has an index between 111.5 and 126.3 in weeks +10 to +15 and 144.3 in week +17. Without an untreated comparison group, such movements cannot be separated from trend growth, seasonality, or changes in the sampled users' activity.

The main specification counts pushes to any classified repository, including repositories first observed after the shutdown. A complementary specification follows only repositories pushed to during the eight baseline weeks (Table A16). Activity in these pre-existing repositories recovers far less: in week +8, their indices range from 10.1 (Accommodation/Food) to 31.4 (Education), compared with 62.6 to 94.5 in the sector-wide specification. Because repositories selected on recent activity would become less active over time even without a shutdown, this gap does not measure a shutdown effect. It is nevertheless consistent with recovery occurring partly through new repositories and switching across projects rather than solely through a return to pre-existing ones.

Taken together, the sectoral evidence indicates that the collapse in observable activity documented in Section 6.1 was followed by a recovery that was gradual and uneven across the economic sectors associated with developers' repositories. These are before–after comparisons within Bangladesh. They share the identification limitations discussed in Section 8.4 and, in addition, cannot separate post-shutdown recovery from other Bangladesh-specific developments between August and November 2024.

The RDiT grid spans −4.45 to −19.52 percentage points for Bangladesh, illustrating substantial specification sensitivity (Table A8). The archived 16-row grid is also retained, but its input and cutoff provenance is unresolved. The RDiT sample includes 174 accounts absent from the main SCM cohort, while 86 SCM accounts are absent from the RDiT; the difference is not simply 88 additional accounts. These facts limit a direct comparison of the two levels and estimates.

7 Mechanism: Offline Work and Delayed Synchronization

7.1 Main backlog measure

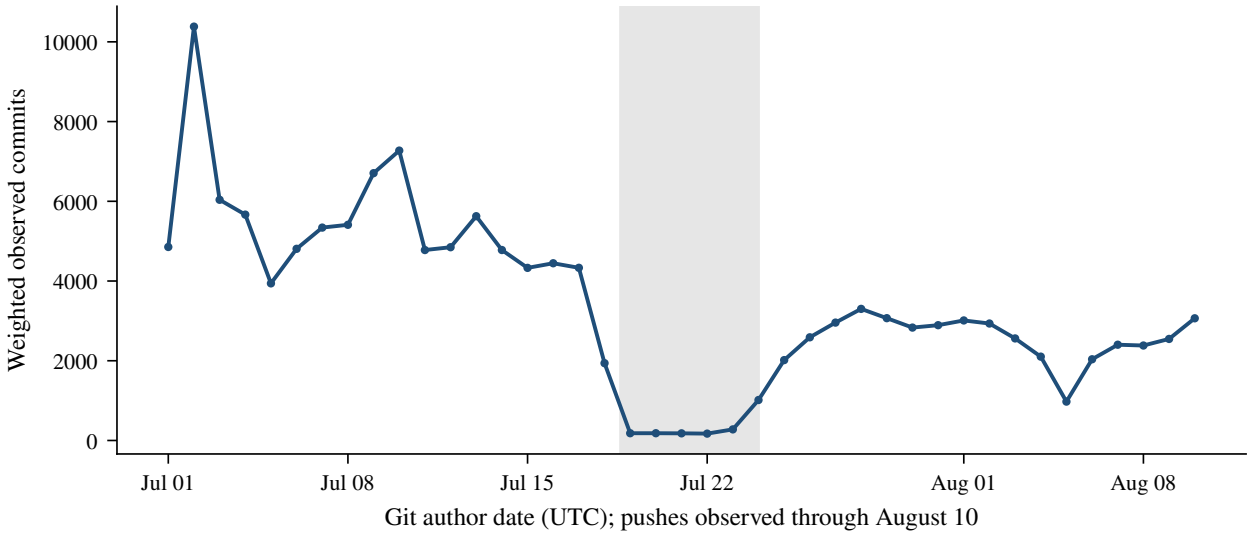
A fall in public GitHub events does not necessarily imply an equal fall in code creation. Git is decentralized: developers may continue editing files and creating commits locally, then push them only after connectivity returns. The commit-level analysis examines this pattern among code that is subsequently observed publicly.

In the strict sample, 2,932 commits are authored during July 18–23. Of these, 829 first appear in a public push on or after July 24. The resulting author-time backlog share is 28.27 percent (Table 9). Replacing the author timestamp with the committer timestamp produces a nearly identical 27.16 percent backlog share (790 of 2,909 commits). The two timestamps are not independent validation clocks: authoredDate and committedDate can be set by the same local client and clock. Their agreement checks sensitivity to the chosen metadata field, not the physical time of coding. Both timestamps are Git metadata generated by the author's local client or the hosting platform at commit time, not an externally verified record of the exact physical moment the code was written; the mechanism evidence is therefore consistent with offline or local work followed by delayed synchronization, but it does not directly observe the underlying production process.

The level comparison substantially limits the adaptation claim. The calendar cohort contains 2,932 commits (835 users), including 1,835 timestamped before 15:00 UTC on July 18, approximately when national connectivity collapsed (Belson, 2024a). Starting the author-time cohort at that onset leaves 1,097 commits from 250 users; 752 were first pushed on or after July 24 (68.55 percent), compared with 829 of 2,932 (28.27 percent) in the original calendar cohort. Both definitions retain pushes observed through August 10.

Inverse push-day sampling weights produce a mean of 33,787.50 commits across the six original six-day placebo windows. The onset cohort is 96.75 percent smaller, but lasts 5.375 rather than six days. Matching the placebo windows to the same 15:00 start gives a baseline of 30,038.26 and a 96.35 percent decline; scaling the six-day baseline by duration gives 96.38 percent. Thus observable creation of code that becomes public within the observation horizon fell roughly 96–97 percent, rather than merely being delayed. The original calendar cohort falls 91.32 percent relative to its six-day baseline. Backlog is small in levels: 829 commits are 2.45 percent of the six-day baseline, and the onset cohort's 752 are 2.50 percent of the matched baseline. These are descriptive level comparisons, not a second causal estimate; the early-July activity increase and unresolved metadata both affect their interpretation.

Figure 6: Observable commits by Git author date: weighted levels



Notes: Strict author-equals-pusher sample, UTC dates; counts weighted by the inverse daily metadata-request probability for pushes before July 18, and weight one thereafter. The date of authorship, not publication, defines the outcome. Zero means no observed qualifying commits. Commits never pushed publicly, pushed after August 10, omitted from incomplete payloads, or unresolved by the metadata query are unobserved; weights correct sampling, not those omissions. Later author cohorts have shorter follow-up.

The 345 onset-cohort commits pushed before July 24 involve 64 users. Of these, 246 were pushed on July 23, when fixed broadband began returning; 99 were pushed earlier. This is incompatible with treating the entire calendar window as complete isolation for every sampled account. Physical location, exceptional access and timestamp error cannot be distinguished locally. Nor is observation complete: 35 calendar-cohort backlog commits first appear on August 5–10. Further pushes after August 10 and code never published are censored. The estimated creation decline therefore applies to code eventually observed publicly within this horizon, not all code written.

Long publication delays become dramatically more common during the shutdown. Among shutdown-authored commits, 29.71 percent have an author-to-push delay above 24 hours, 25.38 per-

cent above 48 hours, and 22.75 percent above 72 hours. In six overlapping pre-shutdown placebo windows, the corresponding averages are approximately 1.10 percent, 0.72 percent, and 0.46 percent (Table A10). Because commit metadata were queried for only a daily sample of commits pushed before July 18 but for all commits pushed from July 18 onward, the placebo rates use inverse-probability weights, while shutdown-authored commits, all pushed on or after July 18, receive weight one. The real shutdown is therefore far outside the normal pre-period delay distribution.

Table 9: Commit-time mechanism

Timestamp	*N* during shutdown	Pushed Jul 24+	Backlog share	>24h	>48h	>72h	Mean delay (h)
`authoredDate`	2,932	829	28.27%	29.71%	25.38%	22.75%	40.94
`committedDate`	2,909	790	27.16%	28.60%	24.27%	21.73%	37.47

Notes: Strict sample requires author login = PushEvent actor, a non-missing timestamp, and a nonnegative delay. The backlog share is conditional on commits observed in the mechanism sample, not a share of all software production in Bangladesh. `authoredDate` and `committedDate` are Git metadata, not externally verified measurements of when code was written.

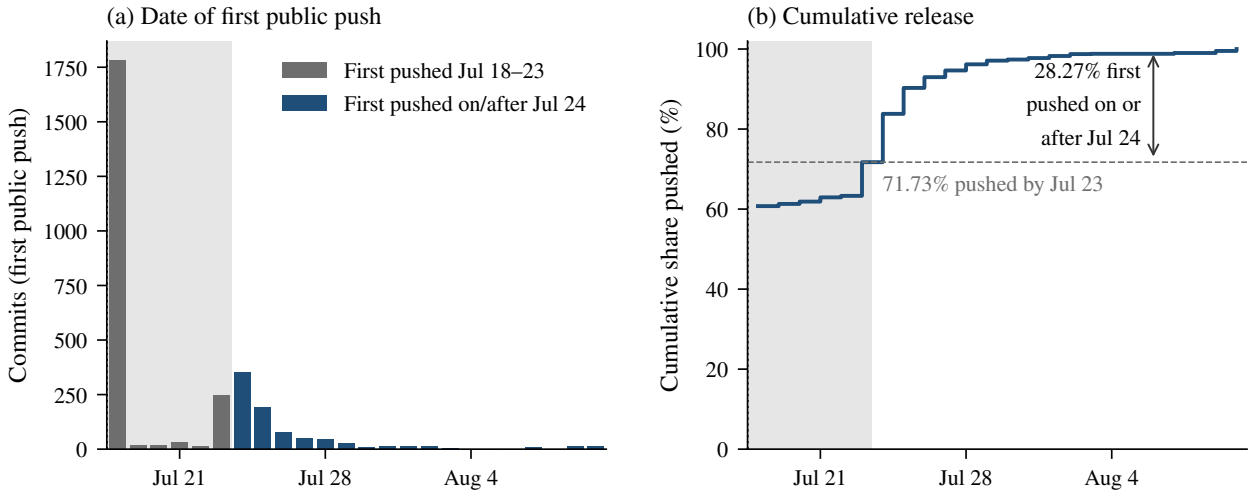
7.2 Timing and concentration of delayed pushes

The delayed commits are released quickly once connectivity begins to return (Figure 7). Of the 829 author-time backlog commits, 354 (42.7 percent) appear on July 24 and 544 (65.6 percent) by July 25. By July 27, 81.1 percent have appeared; by July 31, 92.0 percent have appeared. This temporal concentration immediately after restoration is consistent with a synchronization backlog rather than ordinary long-lived branches or sporadic publication.

The backlog is not generated by only a handful of unusually active users (Table A14). The 2,932 shutdown-authored commits come from 835 users and 977 repositories. Two hundred twenty-two users (26.6 percent) and 246 repositories (25.2 percent) contribute at least one backlog commit. The largest user accounts for 2.77 percent of backlog commits, the top five users for 12.06 percent, and the top ten for 21.47 percent. Dropping the twenty users with the most backlog still leaves a 20.37 percent backlog share.

Backlog is, however, more common in larger pushes (Table A13). Restricting to pushes with at most 10 commits lowers the backlog share to 21.0 percent; at most 5 commits, 14.4 percent; at most 2 commits, 7.8 percent; and single-commit pushes, 3.3 percent. This pattern is economically meaningful rather than merely inconvenient: if developers accumulate local commits while offline and synchronize later, restoration should produce larger-than-normal push batches. At the same time, it means that the 28.3 percent headline mechanism estimate should not be interpreted as applying uniformly to every kind of push.

Figure 7: Release timing of commits with shutdown-period Git author timestamps



Notes: Strict sample: commits whose GitHub-linked author equals the Bangladeshi pusher, with a non-missing author timestamp and nonnegative author-to-push delay; the cohort is the 2,932 commits authored July 18–23 (UTC). Panel (a): date of first public push; panel (b): cumulative share pushed. 829 commits (28.27%) first appear on or after July 24; pushes are observed through August 10. Git `authoredDate` and `committedDate` are metadata, not verified measurements of when code was written; the pattern is consistent with local work followed by delayed synchronization. The share refers only to this strict sample, not to all software production in Bangladesh. Source: Author's calculations using GH Archive and GitHub commit metadata.

8 Robustness, Identification Threats, and Limitations

8.1 Synthetic-control diagnostics

The primary synthetic control improves pre-treatment fit relative to using the Philippines alone, but the fit is not perfect. The pre-treatment RMSPE is 3.26 percentage points, and Bangladesh exhibits temporary deviations from the synthetic series in early July (Figure 1). The international event study shows the same pattern: Bangladesh's pre-shutdown weekly coefficients relative to the donor mean range from −4.89 to 6.10 percentage points and are more volatile than those of any donor (Figure 3). The paper therefore reports pre-fit explicitly rather than treating the synthetic control as a perfect replica. The unusually large post-treatment gap and the stability of the estimate under leave-one-out exercises are the main reasons the design remains informative.

In-time placebos ask whether Bangladesh's own series produces large gaps in ordinary weeks (Abadie et al., 2015). Under a rule fixed in advance—six-day pseudo-treatment windows, non-overlapping and moving backward from July 17, each with at least four weeks of fitting data and with donor weights re-estimated using only earlier days—there are eight eligible pseudo-windows (Figure A1 and Table A5). The actual shutdown ATT of −13.256 percentage points is the largest absolute gap among the nine windows. The largest pseudo-gap is +7.349 percentage points (July 6–11), so the actual ATT is about 1.8 times as large in magnitude and has the opposite sign. The actual

window also ranks first by the RMSPE ratio. Because the pseudo-windows differ in fit length and are not exchangeable, these ranks are descriptive rather than a formal p-value. The three early-July pseudo-windows are all positive, consistent with the elevated pre-shutdown Bangladesh activity visible in the event study (Figure 3).

The same early-July elevation explains why a synthetic difference-in-differences estimator (Arkhangelsky et al., 2021) yields a larger estimate, −17.9495 percentage points (Table A4). The difference comes from the time weights rather than the donor weights: SDID places 72 percent of its time-weight mass on July 1–17, including 27.1 percent on July 2 and 30.2 percent on July 10, so its weighted Bangladesh baseline is 22.54 percent, compared with a uniform pre-period mean of 17.25 percent; the synthetic control and the DiD instead weight all pre-treatment days equally. The comparison therefore shows how sensitive the estimated magnitude is to whether the baseline emphasizes the final pre-shutdown weeks or the full pre-period; it does not indicate that either estimator is invalid. The synthetic control remains the primary specification.

The donor pool also contains countries with their own short-lived connectivity disturbances. Pakistan experienced regional restrictions around July 14–17, immediately before Bangladesh's national shutdown, and Kenya experienced a major traffic disruption on June 25. These episodes are a reason to avoid presenting donor selection as mechanically pristine. Still, removing Pakistan or Kenya individually changes the ATT by less than 0.4 percentage points (Table 10).

Table 10: Primary leave-one-out range

Donor removed	ATT (pp)	Relative effect	Pre RMSPE	RMSPE ratio
Ghana	−13.209	−86.73%	3.297	4.131
Pakistan	−12.905	−86.46%	3.376	3.961
Kenya	−12.886	−86.44%	3.293	4.042
Any zero-weight donor (11)	−13.256	−86.77%	3.260	4.192

Notes: Removing donors that receive zero baseline weight mechanically leaves the baseline estimate unchanged.

A more conservative future extension could pre-register donor exclusion rules based on third-party connectivity data and expand the donor pool to increase placebo-inference resolution.

8.2 Metadata and measurement

Commit metadata resolution is similar across periods: 82.34 percent of the pre-period target commits are found, compared with 84.24 percent during the shutdown and 81.81 percent after (Table A11). The shutdown mechanism is therefore not generated by a sharp deterioration in commit-object availability during the treatment window.

A related limitation concerns push-listing completeness: although 99.5–99.8 percent of pushes

in every period expose a complete commit list, the share of nominal commits captured within complete pushes falls from 90.27 percent before the shutdown to 73.11 percent during the shutdown and 71.45 percent afterward (Table A12). This occurs because a small number of large pushes account for a disproportionate share of commits and are more likely to have incomplete commit listings. Since the strict mechanism sample restricts to complete pushes, this does not mechanically invalidate the 28.27 percent backlog estimate, but it does imply uneven commit-level coverage across periods. The lower coverage during and after the shutdown is therefore a measurement limitation that should be considered when interpreting the mechanism results.

Location is another measurement limitation. GitHub location fields are user-entered and may be missing, stale, or imprecise. The estimates should be read as applying to accounts associated with Bangladesh and the donor countries, not as a verified census of developers physically located in each country on every date. Public-event data additionally exclude private repositories. If private and public work respond differently to a shutdown, the paper estimates the effect on the observable public margin rather than on all software production.

8.3 Sector classification and the sectoral recovery analysis

Table A15 repeats the sectoral recovery analysis keeping only repositories whose classifier confidence is at least 0.70 or 0.90 instead of 0.80. The qualitative ordering of the main sectors is stable. At all three thresholds, Finance and Information begin their first run of two consecutive weeks at or above 80 in week +3; Education and Health do so between weeks +5 and +8; and Accommodation/Food never does, with week +8 indices between 60.3 and 64.9. Real Estate's late recovery is broadly but not fully robust: its week +8 index is 90.0 and 85.5 at the 0.70 and 0.80 thresholds but 75.5 at 0.90. Retail's week +4 overshoot appears at every threshold (indices between 153.7 and 159.1), but the start of its first two-week run at or above 80 varies from week +4 (0.70) to week +9 (0.80) and week +7 (0.90). I therefore do not emphasize a precise recovery date for Retail.

Three sectors with substantial activity are excluded from the main results. Professional/ Scientific and Administrative Support have 577 and 196 sampled users active in the baseline period at the 0.80 threshold, but only 125 and 48 at 0.90, so their indices depend heavily on the threshold. In Arts, Entertainment, and Recreation, a single repository accounts for 82.5 percent of pre-shutdown events among the sector's classified repositories, so a sector index would largely track one project.

The classification has further limitations. First, the sector describes the application domain inferred from a repository's text, not the employer or industry of the developer: a developer employed in banking may work on a general-purpose library, and a student exercise may be classified as Retail because it implements an online shop. Second, classifier errors are possible, and this paper does

not validate the classifier's accuracy on Bangladeshi repositories. The model card reports that the training labels are GPT-4.1 judgements rather than human annotations and that, on public GitHub repositories, roughly half of what the model labels confidently arguably belongs to no sector at all, such as coursework, portfolios, exercises, and generic tooling (Quispe and Xu, 2026); the score threshold mitigates but does not eliminate this. Third, coverage is incomplete. Only 55.1 percent of repositories have a recoverable historical README, and activity in unclassified repositories does not enter any sector. If newly created repositories are less likely than established ones to have a README at their first observed activity date, post-shutdown recovery through new projects would be understated. Fourth, a user may contribute to several sectors, so sector indices are not shares of a fixed workforce. Finally, the sectoral confidence intervals do not adjust for the many sector-week comparisons reported, and they reflect sampling variation across users rather than common shocks.

8.4 Concurrent unrest and causal interpretation

The most important identification threat is that the Internet shutdown was itself a response to political unrest. Protests, curfews, transportation disruptions, workplace closures, fear, or other domestic conditions could directly reduce software work even if connectivity were available. Comparisons with donor countries, whether through synthetic-control weights or the equal-weighted DiD, cannot absorb shocks unique to Bangladesh that coincide with the shutdown. For this reason, the headline estimate should be interpreted as the effect of the shutdown episode under the identifying assumption that the abrupt Bangladesh-specific deviation is primarily attributable to lost connectivity.

Several findings make the connectivity interpretation more plausible without fully proving it. First, the break is tightly aligned with the network outage. Second, activity begins to recover as connectivity returns (Figures 3 and 5). Third, downstream online tasks such as shipping and collaboration fall at least as strongly as the general activity rate. Fourth, commit metadata reveal a large stock of code authored during the outage but pushed immediately after restoration. A pure 'developers stopped working because of unrest' story does not naturally predict that last pattern. Nevertheless, the mechanism evidence is complementary, not a substitute for randomized assignment.

Selection on at least four pre-period event days can mechanically favor a temporarily active cohort and induce subsequent mean reversion. The equally weighted donor mean falls by 2.146 percentage points between the pre-period and July 18–23. This decline is compatible with selection and common time effects; it does not separately identify mean reversion. The same eligibility rule across countries mitigates but does not eliminate differential selection. Diagnostics that fix eligibility before the estimation period are prepared for advisor review, with the original sample retained in the main results.

Pakistan, which receives 47.6 percent of the original synthetic weight, experienced a national connectivity disruption on July 31 (about 08:30–10:30 UTC; Belson, 2024a). This occurs after the primary treatment window but within the recovery comparison. It can depress the synthetic post-period series and inflate Bangladesh's apparent rebound. The primary July 18–23 estimate is unaffected by that date; extrapolated recovery is less secure.

9 Discussion

The results illustrate why high-frequency digital trace data can add information beyond aggregate estimates of shutdown costs. A national GDP series cannot show whether digital workers cease production, shift to offline tasks, or accumulate work that is released later. GitHub events and metadata measure observed public activity and delayed publication; they cannot observe work that never becomes public or distinguish all unobserved adaptation from lost production.

The main estimate—an 86.8 percent decline in the daily active-user rate relative to the synthetic counterfactual (Table 4)—is best interpreted as a collapse in observable online software-production activity. The mechanism shows a small observed backlog alongside a collapse in author-time levels: creation of code later observed publicly falls roughly 96–97 percent from the pre-shutdown comparison, and backlog represents about 2.5 percent of normal volume. The original 28.27 percent backlog share conditions on a much smaller calendar-cohort denominator. It does not establish that missing activity was displaced across time, or quantify destruction of work that never became public.

At the same time, delayed synchronization cannot account for the entire collapse. Most eligible users record no productive public activity during the shutdown, and the recovery backlog is much smaller than the counterfactual volume implied by the active-rate and shipping estimates. Connectivity therefore appears to matter not only for publishing already-written code but also for complementary tasks: accessing remote dependencies, coordinating with collaborators, reviewing code, resolving issues, merging pull requests, and releasing software. The especially large declines in shipping and collaboration are consistent with these channels.

The sectoral results add a dynamic dimension. In aggregate, Bangladesh's gap relative to the donor mean returned within its pre-shutdown range in the first week after restoration (Figure 3), but that wide and volatile range is a weak recovery criterion. The separate daily, rolling-mean and cumulative measures in Section 6 give different return dates and do not show full compensation by September 4. Reconnection did not restore observable activity at once: weeks after the shutdown, activity in Education-, Health-, and especially Accommodation/Food-related repositories remained below its pre-shutdown level, while Finance- and Information-related activity recovered within about three weeks. This pattern is consistent with the costs of a shutdown extending beyond

the days without connectivity and falling unevenly across types of projects. The weaker recovery of pre-existing repositories, relative to sector-wide activity, further suggests that part of the recovery took the form of new projects or switching across repositories, which an analysis restricted to pre-existing projects would miss. Because the sectoral analysis has no untreated comparison group and classifies repositories rather than workers, these interpretations remain suggestive.

The national cumulative comparison is more informative about recovery than a first return inside a wide pre-period range. Relative to the unchanged synthetic series, 8,145 active user-days are missing in the six-day shutdown window; later positive gaps partly offset, but never fully repay, the cumulative shortfall, which is 3,633 user-days at September 4. Neither this arithmetic offset nor sectoral baseline crossings identify a causal rebound. The data do not evaluate particular resilience policies, so no comparative policy recommendation is drawn.

10 Conclusion

Bangladesh's July 2024 Internet shutdown provides a high-frequency case study of what happens to digitally mediated production when national connectivity is abruptly removed. Using public GitHub events for 59,158 pre-active accounts across fifteen countries, the paper finds that Bangladesh's daily productive-activity rate fell by 13.26 percentage points relative to a synthetic counterfactual during July 18–23, an 86.8 percent decline. International difference-in-differences, secondary outcomes, placebos, and temporal discontinuity analyses point in the same direction (Table 3). The cumulative synthetic-control deficit is 8,145 active user-days in the shutdown window and 3,633 by September 4 after partial offsets. A first daily crossing on July 27 is weaker evidence of recovery than the first nonnegative seven-day mean gap on August 13. These descriptive sums inherit imperfect pre-fit and concurrent-shock limitations.

The commit-level evidence shows that the observed backlog was small relative to normal creation. After the approximately 15:00 UTC July 18 onset, 1,097 commits were authored and subsequently observed publicly by August 10, a 96.35 percent fall against duration-matched pre-shutdown windows (96.75 percent against the original six-day windows). Backlog is about 2.5 percent of normal volume. The original calendar-cohort backlog share of 28.27 percent and onset-cohort share of 68.55 percent describe delayed publication among the surviving observed commits; neither establishes that total software work continued at a comparable rate.

A within-Bangladesh analysis of the restoration period adds a further qualification: reconnection did not immediately restore observable activity. Recovery was faster in Finance- and Information-related repositories than in Education- and Health-related repositories, and activity in Accommodation/Food-related repositories remained depressed throughout the observed window. These sectoral patterns are descriptive comparisons with each sector's own pre-shutdown level

rather than causal estimates against an untreated control.

The paper's main limitation is the coincidence of the shutdown with severe domestic unrest. Future work could combine network telemetry at subnational or provider level with developer location at finer geographic resolution, or exploit variation in restoration timing across networks and regions. Such designs could more sharply separate the direct effect of connectivity from the broader social environment. Even with this limitation, the evidence is consistent with observable software-development activity being highly exposed to network availability and with public development platforms providing unusually granular measures of that exposure.

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A Additional Empirical Diagnostics

This appendix reports the detailed tables behind the synthetic-control, difference-in-differences, mechanism, and sectoral recovery results discussed in the main text. All tables are generated from the stored result files by `analysis/tables/make_all_tables.py`.

A.1 Synthetic control

Table A1: Primary synthetic-control donor weights

Country	Weight
Pakistan	47.579%
Ghana	32.488%
Kenya	19.933%

Notes: Weights are estimated on the pre-treatment (May 1–July 17, 2024) daily active-user rate. The remaining eleven donor countries receive zero weight.

Table A2: Synthetic-control in-space placebo inference

Rank	Unit	ATT (pp)	Pre RMSPE	Post RMSPE	Ratio
1	**Bangladesh**	−13.256	3.260	13.666	4.192
2	Tunisia	−2.563	2.011	2.682	1.334
3	Nepal	0.965	0.936	1.216	1.300
4	Jordan	−0.194	2.457	3.151	1.283
5	Indonesia	−1.393	1.151	1.462	1.270
6	Pakistan	0.941	1.585	1.606	1.013
7	Kenya	2.439	2.873	2.837	0.987
8	Nigeria	0.527	0.948	0.823	0.869
9	Sri Lanka	−0.748	1.382	1.181	0.855
10	Thailand	−0.215	0.965	0.652	0.675
11	Ghana	0.268	1.144	0.706	0.617
12	Vietnam	−0.097	0.867	0.458	0.528
13	Philippines	0.091	0.955	0.416	0.435
14	Morocco	0.050	1.794	0.625	0.348
15	Malaysia	0.131	1.177	0.297	0.252

Notes: Each donor country is assigned the July 18–23 treatment in turn and its synthetic control is rebuilt from the remaining donors. Units are ordered by the post/pre RMSPE ratio. Bangladesh has both the largest ratio and the largest absolute ATT among the 15 units, so the exact rank p-value is $1/15 = 0.0667$ for both statistics. This is the minimum attainable value with 14 placebo countries; inference resolution is limited by the size of the donor pool.

Table A3: Outcome-specific synthetic-control donor weights

Outcome	Country	Weight
Productive active rate	Pakistan	30.0%
	Ghana	26.1%
	Philippines	23.8%
	Kenya	18.0%
	Morocco	2.2%
Shipping active rate	Pakistan	72.4%
	Morocco	10.8%
	Ghana	9.2%
	Malaysia	4.8%
	Jordan	2.4%
	Tunisia	0.5%
Collaboration active rate	Pakistan	73.6%
	Morocco	26.4%
Collaboration events / 1,000	Morocco	84.0%
	Indonesia	13.5%
	Pakistan	2.5%
Shipping output / 1,000	Pakistan	57.2%
	Morocco	17.5%
	Ghana	8.9%
	Nigeria	8.1%
	Nepal	4.1%
	Malaysia	1.8%
	Jordan	1.4%
	Tunisia	0.7%
	Philippines	0.2%

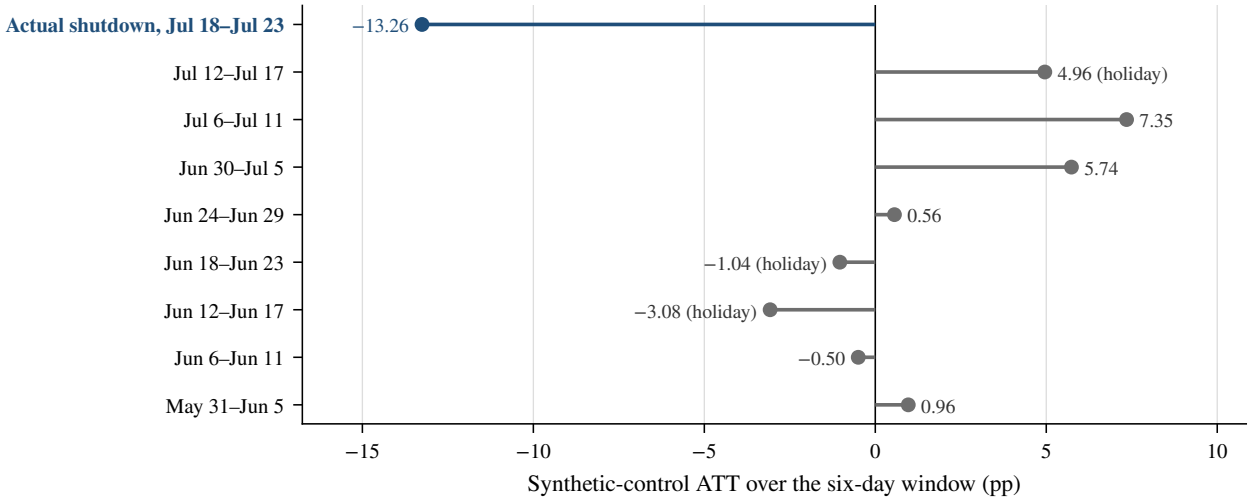
Notes: Donors with zero weight are omitted. These weights are reported for transparency; the main secondary-outcome comparison holds the primary weights in Table A1 fixed.

Table A4: Synthetic difference-in-differences: sensitivity to the pre-treatment baseline

Statistic	Value	
A. Estimates, July 18–23, 2024 (pp)		
Synthetic DiD (archived)	−17.9495	–
Canonical synthetic control (primary)	−13.256	–
B. Decomposition by weighting scheme (pp)		
Unit weights	SDID time weights	Uniform time weights
SDID unit weights	−17.95	−13.86
SC unit weights	−17.99	−14.38
Equal unit weights	−17.87	−13.09
C. SDID time weights		
Share of weight on July 1–17	72.2%	
Weight on July 2 / July 10	27.1% / 30.2%	
Days with positive weight (of 78)	14	
Bangladesh baseline, SDID-weighted (%)	22.54	
Bangladesh baseline, uniform pre-period mean (%)	17.25	

Notes: Panel A: the archived SDID estimate (Arkhangelsky et al., 2021) and the primary synthetic control. The archived SDID standard error and p-value are not reported because their computation is undocumented. Panel B recomputes the point estimate from the archived SDID unit and time weights, replacing one set at a time; equal unit weights with uniform time weights reproduce the aggregate DiD. Panel C summarizes the archived time weights over May 1–July 17. Nothing is re-estimated. Sources: `results/synthetic_control/main_synthetic_results.csv`, `sdid_time_weights.csv`, `donor_weights.csv`; computed by `analysis/synthetic_control/02_sdid_archived_check.py`.

Figure A1: Synthetic-control in-time placebos: ATT over six-day pseudo-treatment windows



Notes: Each grey row is a six-day pre-treatment window chosen by the rule in Section 8.1; donor weights are re-estimated using only the days before each window, and the ATT is the mean gap between Bangladesh and its synthetic control over the window, as in the primary specification (blue row). "Holiday" marks windows containing a Bangladesh national holiday. All eight eligible windows are shown. Details in Table A5. Source: `results/synthetic_control/reproduced/sc_intime_placebos.csv`.

Table A5: Synthetic-control in-time placebos

Window (2024)	Fit days	ATT (pp)	Pre RMSPE	Ratio	Rank |ATT|	Rank ratio	Holiday
Jul 18–Jul 23 (actual)	78	−13.256	3.260	4.192	1	1	
Jul 12–Jul 17	72	4.958	3.032	1.787	4	5	Ashura
Jul 6–Jul 11	66	7.349	2.279	3.376	2	3	
Jun 30–Jul 5	60	5.736	1.634	3.793	3	2	
Jun 24–Jun 29	54	0.560	1.685	0.654	8	9	
Jun 18–Jun 23	48	−1.036	1.678	1.227	6	6	Eid al-Adha
Jun 12–Jun 17	42	−3.076	1.546	2.487	5	4	Eid al-Adha
Jun 6–Jun 11	36	−0.498	1.588	0.844	9	7	
May 31–Jun 5	30	0.964	1.683	0.720	7	8	

Notes: Pre-specified rule: six-day pseudo-treatment windows, non-overlapping, moving backward from July 17, each with at least 28 days of fitting data from May 1. Donor weights are re-estimated for each window using only days before it; ATT and RMSPE definitions are those of the primary specification. Ranks are among the nine windows. The windows differ in fit length and are not exchangeable, so the ranks are descriptive and not a formal p-value. Source: `results/synthetic_control/reproduced/sc_intime_placebos.csv`, computed by `analysis/synthetic_control/03_intime_placebos.py`.

A.2 Difference-in-differences and event study

Table A6: Aggregate event study: all weekly estimates

Week	Dates (2024)	Days	$\hat{\theta}_k$ (pp)	Placebo-based interval	Placebo range	Rank p
−11	May 1–May 8	8	−0.86	[−4.37, 2.65]	[−1.68, 4.62]	0.6000
−10	May 9–May 15	7	−1.83	[−4.50, 0.84]	[−2.23, 2.49]	0.2667
−9	May 16–May 22	7	0.58	[−1.21, 2.36]	[−1.06, 2.46]	0.4667
−8	May 23–May 29	7	−0.51	[−2.47, 1.45]	[−2.36, 1.92]	0.4667
−7	May 30–Jun 5	7	−0.05	[−2.08, 1.97]	[−1.72, 2.02]	1.0000
−6	Jun 6–Jun 12	7	−1.23	[−3.28, 0.81]	[−0.99, 2.54]	0.2000
−5	Jun 13–Jun 19	7	−4.89	[−9.31, −0.47]	[−3.64, 2.34]	0.0667
−4	Jun 20–Jun 26	7	−2.71	[−5.13, −0.29]	[−3.06, 1.62]	0.1333
−3	Jun 27–Jul 3	7	1.67	[−0.46, 3.80]	[−1.31, 1.45]	0.0667
−2	Jul 4–Jul 10	7	6.10	[3.76, 8.43]	[−1.74, 1.62]	0.0667
−1	Jul 11–Jul 17	7	3.86	[2.15, 5.58]	[−1.63, 0.98]	0.0667
0	Jul 18–Jul 23	6	−13.09	[−15.39, −10.79]	[−1.72, 1.60]	0.0667
+1	Jul 24–Jul 30	7	−2.66	[−5.35, 0.02]	[−1.95, 2.23]	0.0667
+2	Jul 31–Aug 6	7	−2.76	[−4.87, −0.65]	[−1.22, 2.56]	0.0667
+3	Aug 7–Aug 13	7	0.52	[−1.99, 3.02]	[−1.87, 2.73]	0.9333
+4	Aug 14–Aug 20	7	4.31	[1.99, 6.64]	[−1.82, 2.77]	0.0667
+5	Aug 21–Aug 27	7	4.84	[2.40, 7.28]	[−1.48, 2.71]	0.0667
+6	Aug 28–Sep 3	7	2.53	[0.01, 5.05]	[−2.32, 2.31]	0.0667

Notes: $\hat{\theta}_k$ from Equation (3), relative to the May 1–July 17 mean. Week −11 includes May 1; September 4 (a partial week) is omitted. *Placebo range*: minimum and maximum of the 14 in-space placebo estimates. *Placebo-based interval*: $\hat{\theta}_k \pm 1.96$ times the placebo standard deviation; a donor-placebo reference interval, not a conventional confidence interval. Other definitions as in Table 5. Source: `results/did/did_eventstudy_weekly.csv`.

Table A7: International difference-in-differences by outcome

	All days		Holidays excluded	
Outcome	ATT	Relative	ATT	Relative
Active-user rate (primary)	−13.088	−86.6%	−13.379	−86.9%
Productive active rate	−11.748	−86.7%	−12.007	−86.9%
Shipping active rate	−0.565	−89.0%	−0.590	−89.4%
Collaboration active rate	−0.266	−86.3%	−0.269	−86.4%
Collaboration events / 1,000	−10.575	−91.8%	−10.685	−91.9%
Shipping output / 1,000	−10.380	−88.5%	−10.827	−88.9%

Notes: Aggregate DiD as in Table 5, Panel A, for each outcome; rates in percentage points, counts per 1,000 eligible users. No placebo inference is stored for these specifications. Source: `results/final/table_DiD_all_outcomes_eligible.csv`.

A.3 Regression discontinuity in time

Table A8: RDiT: complete cutoff and bandwidth grid

Cutoff	h	Bangladesh	BD minus PH	Philippines
07-17	4	-4.45 (3.45)	-2.40 (3.43)	-2.05 (0.53)
07-17	5	-7.30 (4.36)	-6.56 (4.87)	-0.74 (1.54)
07-17	7	-9.28 (5.03)	-9.16 (4.98)	-0.12 (1.49)
07-17	10	-13.70 (6.05)	-13.07 (5.80)	-0.62 (1.35)
07-17	14	-16.80 (5.64)	-16.04 (5.35)	-0.76 (1.16)
07-18	4	-11.09 (2.72)	-7.47 (2.14)	-3.62 (2.01)
07-18	5	-12.98 (3.12)	-8.75 (1.94)	-4.24 (1.86)
07-18	7	-17.49 (4.48)	-15.58 (4.29)	-1.91 (1.34)
07-18	10	-18.06 (3.94)	-17.35 (4.18)	-0.71 (1.31)
07-18	14	-19.52 (3.47)	-17.82 (3.50)	-1.70 (1.07)
07-19	4	-9.00 (2.49)	-6.69 (2.12)	-2.31 (1.44)
07-19	5	-11.54 (2.91)	-7.73 (1.99)	-3.81 (1.84)
07-19	7	-16.59 (4.24)	-13.80 (4.06)	-2.79 (1.19)
07-19	10	-16.76 (3.15)	-15.52 (3.59)	-1.23 (1.17)
07-19	14	-17.60 (2.99)	-15.50 (3.16)	-2.10 (1.00)

Notes: All effects and HAC(2) standard errors in percentage points. Local linear fits with separate slopes, cutoff day on the right, inclusive symmetric bandwidth, $n/(n-4)$ variance correction. All 45 fits are reported; none replaces the archived July 17, $h = 7$ result. Source: new round-2 diagnostic grid.

Table A9: RDiT: preserved archived 16-specification grid

Pre bandwidth	Polynomial	Donut	Effect (pp)	SE (pp)	*n*
7	lin	False	-14.0	2.2	14
7	lin	True	-20.4	1.5	12
7	quad pre	False	-16.1	2.8	14
7	quad pre	True	-21.2	2.5	12
10	lin	False	-13.1	1.9	17
10	lin	True	-18.5	1.0	15
10	quad pre	False	-15.2	2.7	17
10	quad pre	True	-20.5	1.7	15
14	lin	False	-14.8	2.4	21
14	lin	True	-20.6	1.5	19
14	quad pre	False	-13.2	2.3	21
14	quad pre	True	-18.1	1.4	19
21	lin	False	-17.7	3.2	28
21	lin	True	-23.7	2.5	26
21	quad pre	False	-12.4	2.1	28
21	quad pre	True	-17.5	1.2	26

Notes: Transcribed without re-estimation from data/processed/table3_rdit_sensitivity.csv. The historical cutoff, outcome, sample and variance procedure are not documented. The row labelled linear, 7 days, no donut is -14.0 pp, compared with -9.28 pp in the reproduced preferred specification; this is an unresolved provenance conflict, not evidence that one specification should replace the other.

A.4 Mechanism

Table A10: Mechanism placebo comparison

Metric	Real shutdown	Mean placebo	Maximum placebo
Released in next 6 days	25.38%	0.56%	0.77%
Delay >24h	29.71%	1.10%	1.31%
Delay >48h	25.38%	0.72%	0.90%
Delay >72h	22.75%	0.46%	0.56%
Delay >120h	13.88%	0.32%	0.46%

Notes: Placebo windows are six six-day authoring windows between July 1 and July 11, with six-day subsequent release windows that end before treatment. Windows overlap, so the comparison is descriptive rather than an exact independent randomization test. Commit metadata were queried for all first-push commits pushed on or after July 18 but for a daily sample of those pushed July 1–17 (7–17 percent of commits per push date). Placebo rates therefore weight each commit by the inverse of its push date's query rate (number of first-push commits divided by number queried); commits authored July 18–23 can only be pushed on or after July 18 and receive weight one, so all rows estimate the same population share. Without weights, placebo rates would over-represent late pushes. Mean placebo N is 4,144 observed commits per window. Sources: `results/final/table_mechanism_placebos.csv`, reproduced by `analysis/mechanism/01_placebo_weighting_audit.py`.

Table A11: Metadata coverage by push period

Period	Target N	Commit found	Author timestamp	Author login	Author = pusher
Pre	17,000	82.34%	82.34%	77.95%	71.49%
Shutdown	3,197	84.24%	84.24%	79.17%	69.69%
Post	65,868	81.81%	81.81%	77.81%	71.05%

Notes: Similar resolution rates across periods reduce concern that the mechanism result is generated by treatment-specific metadata missingness.

Table A12: Push-listing completeness by period

Period	Pushes	Mean push size	Complete pushes	Nominal commits	Commits in complete pushes
Pre	121,960	1.29	99.78%	157,834	90.27%
Shutdown	2,738	1.70	99.53%	4,667	73.11%
Post	58,110	1.69	99.66%	98,022	71.45%

Notes: A push is complete when its historical payload lists every commit. Large pushes account for a disproportionate share of commits and are more likely to be incomplete. Source: `results/mechanism/robust_push_size_period.csv`.

Table A13: Mechanism estimates by push size

Sample	N	Backlog N	Backlog share	>24h	>48h	>72h
All complete pushes	2,932	829	28.27%	29.71%	25.38%	22.75%
Push size ≤ 10	2,503	526	21.01%	21.81%	18.98%	17.14%
Push size ≤ 5	2,217	320	14.43%	15.07%	13.62%	12.45%
Push size ≤ 2	1,947	151	7.76%	7.96%	7.29%	6.93%
Single-commit pushes	1,707	57	3.34%	3.28%	2.93%	2.75%

Notes: Strict `authoredDate` sample of commits authored July 18–23. The backlog share is not invariant to push size.

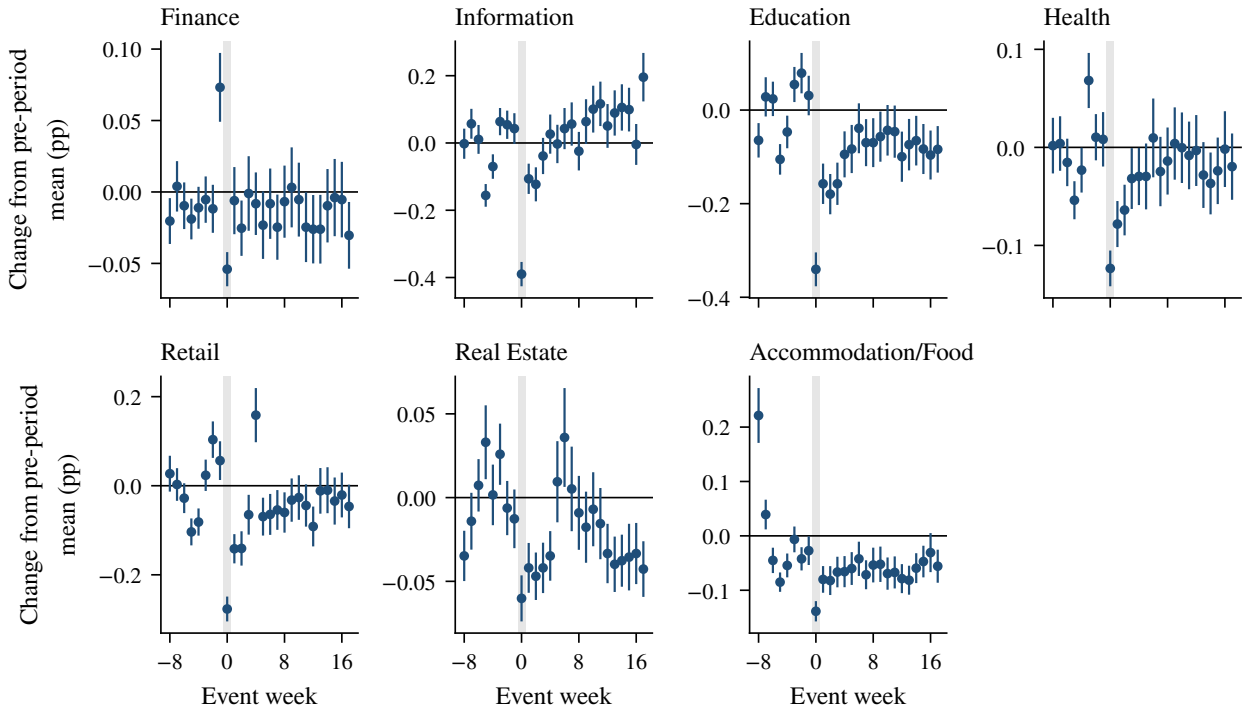
Table A14: Concentration of the backlog across users and repositories

Statistic	Value
Shutdown-authored commits	2,932
Backlog commits	829 (28.27%)
Users / with backlog	835 / 222 (26.59%)
Repositories / with backlog	977 / 246 (25.18%)
Largest user's share of backlog	2.77%
Top 5 users' share of backlog	12.06%
Top 10 users' share of backlog	21.47%
Backlog share after dropping top-k backlog users	
$k = 1$ ($N = 2,909$)	27.71%
$k = 3$ ($N = 2,868$)	26.67%
$k = 5$ ($N = 2,831$)	25.75%
$k = 10$ ($N = 2,739$)	23.77%
$k = 20$ ($N = 2,577$)	20.37%

Notes: Strict `authoredDate` sample. Users are ranked by their number of backlog commits, breaking ties by lowercase login in ascending order. The deterministic audit reproduces the archived top-10 exclusion ($N = 2,739$, 23.77 percent); the referee reports $N = 2,740$ and 23.76 percent with an unspecified tie order. Both figures are recorded as a tie-handling discrepancy; the archived result is preserved.

A.5 Sectoral recovery

Figure A2: Sectoral event study: change in daily push-activity probability relative to the pre-shutdown baseline



Notes: Points are β_{sk} from Equation (5): the mean change, across the fixed sample of 20,000 Bangladesh-associated accounts, in the daily probability of a PushEvent in a sector-s repository during event week k, relative to each user's mean over May 23–July 17, 2024, in percentage points. Vertical lines are 95 percent confidence intervals from the cross-user standard deviation of these differences; they capture sampling variation across users but not common weekly shocks. Shaded: July 18–23 shutdown; the window ends at week +17 (November 13–19). Vertical scales differ across panels. Within-Bangladesh descriptive comparison, not a causal effect relative to an untreated control. Source: Author's calculations using GH Archive.

Table A15: Sectoral recovery indices by classifier-confidence threshold

Sector	Threshold	Pre users	Week +1	Week +4	Week +8	First two-week run ≥80
Finance	0.70	437	91.6	92.4	89.2	3
	0.80	382	92.2	89.5	91.3	3
	0.90	275	101.7	82.5	101.7	3
Information	0.70	2,123	73.0	116.6	101.3	3
	0.80	1,528	75.9	106.0	94.5	3
	0.90	968	79.5	97.6	91.5	3
Education	0.70	1,345	61.6	75.4	79.4	5
	0.80	1,234	59.1	75.4	81.9	6
	0.90	1,029	57.8	73.3	77.0	6
Health	0.70	542	40.8	76.4	88.3	6
	0.80	514	40.7	77.5	89.4	6
	0.90	455	41.1	75.5	87.6	8
Retail	0.70	1,090	52.2	159.1	82.5	4
	0.80	1,012	52.0	153.7	79.6	9
	0.90	829	50.1	157.4	80.9	7
Real Estate	0.70	228	33.1	45.5	90.0	5
	0.80	213	33.0	44.4	85.5	5
	0.90	153	44.9	44.9	75.5	5
Accommodation/Food	0.70	514	46.0	59.3	64.9	–
	0.80	480	44.3	54.2	62.6	–
	0.90	394	39.2	45.1	60.3	–
Excluded from the main figures and table						
Professional/Scientific	0.70	1,241	57.9	66.9	91.2	7
	0.80	577	64.9	64.9	91.0	7
	0.90	125	94.4	65.6	78.7	1
Admin. Support	0.70	268	30.8	83.0	77.0	9
	0.80	196	32.2	72.9	84.7	10
	0.90	48	5.2	61.9	56.8	–

Notes: Each row repeats the main specification (Table 8) keeping only repositories whose classifier confidence is at least the stated threshold; 0.80 is the main specification. Professional/Scientific and Administrative Support are excluded from the main results because their support falls sharply at the 0.90 threshold. Source: `results/sector_recovery/final/sector_threshold_robustness_full.csv`.

Table A16: Recovery of pre-existing repositories versus sector-wide user activity

		Fixed pre-period repositories			Sector-wide (main)		
Sector	Repos	+1	+4	+8	+1	+4	+8
Finance	417	60.0	28.3	23.0	92.2	89.5	91.3
Information	2,040	46.8	37.4	27.3	75.9	106.0	94.5
Education	1,523	40.3	38.2	31.4	59.1	75.4	81.9
Health	651	25.1	28.6	23.1	40.7	77.5	89.4
Retail	1,322	28.7	30.9	19.8	52.0	153.7	79.6
Real Estate	254	16.1	25.7	30.0	33.0	44.4	85.5
Accommodation/Food	581	29.4	16.1	10.1	44.3	54.2	62.6

Notes: Recovery indices (pre-period mean = 100). *Fixed pre-period repositories*: classified repositories (confidence ≥ 0.80) with at least one PushEvent by a sampled user during the final eight pre-shutdown weeks; the outcome is each repository's daily PushEvent active-day rate, and new repositories never enter. *Sector-wide*: main specification of Table 8, which includes activity on repositories first observed after the shutdown. Because a cohort selected on recent activity would decline over time even without a shutdown, the fixed-repository indices are not a counterfactual for the sector-wide indices. Sources: `results/sector_recovery/pre8/sector_recovery_summary_pre8.csv` and `results/sector_recovery/sectorwide/sectorwide_recovery_summary.csv`.